Security Lab Solution Document

Cloud IaaS Integration – AWS, Azure and GCP

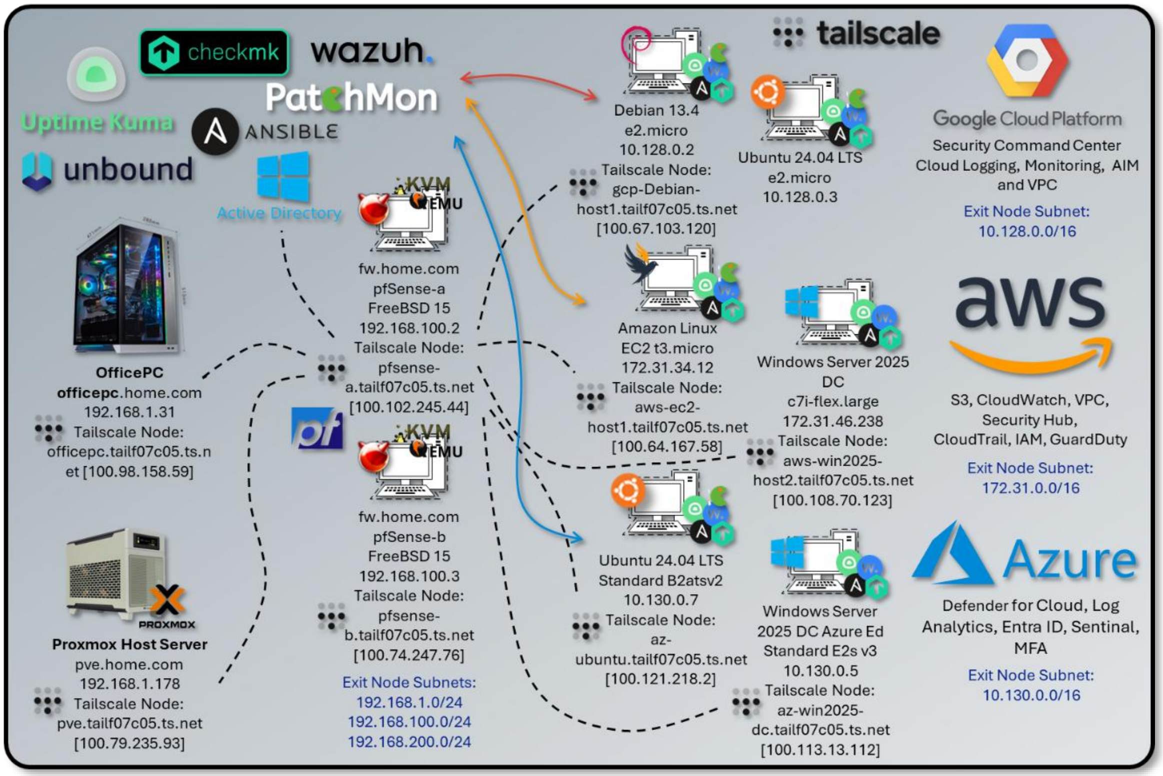
Created By: Paul Leone

March 31, 2026

Contents

Cloud IaaS Integration Overview

Architecture Overview



The lab extends beyond the on-premises Proxmox environment into a multi-cloud hybrid architecture spanning AWS, Azure, and Google Cloud Platform. Tailscale provides the encrypted mesh VPN fabric connecting all cloud-hosted nodes back to on-premises infrastructure without exposing management interfaces to the public internet. Each cloud provider hosts purpose-built workloads aligned with their native security services, enabling hands-on experience with enterprise cloud security patterns across the three major platforms.

All cloud nodes integrate with a common on-premises management stack — Wazuh EDR, PatchMon, Checkmk, Ansible, and Uptime Kuma — configured consistently across providers. DNS resolution uses the lab Unbound resolvers (192.168.1.153 / 192.168.1.154) accessed via Tailscale. Full integration details are covered in the On-Premises Management Integration section.

Platform	VM / Instance	OS	Primary Services	Tailscale Node
AWS	EC2 t3.micro EC2 c7i-flex.large	Amazon Linux 2 Windows Server 2025 Datacenter	S3, CloudWatch, VPC, Security Hub, CloudTrail, IAM, Lambda, GuardDuty	aws-ec2-host1.tailf07c05.ts.net [100.64.167.58] aws-win2025-host2.tailf07c05.ts.net [100.108.70.123]
GCP	e2.micro (x2)	Debian 13.4 Ubuntu	Security Command Center, Cloud Logging, Cloud Monitoring, IAM, VPC, Cloud Armor	gcp-debian-host1.tailf07c05.ts.net [100.67.103.120]
Azure	Standard_ B2ats v2 Standard_ E2s v3 (Spot)	Ubuntu 24.04 LTS Windows Server 2025 Datacenter Smalldisk	Defender for Cloud, Log Analytics, Entra ID, Sentinel, MFA, NSG, Azure Arc, DDoS Protection	az-ubuntu.tailf07c05.ts.net [100.96.110.40] az-win2025-

Security Impact

- Cross-cloud EDR coverage via Wazuh agents reporting to the central on-premises manager at 192.168.1.219
- Cloud-native security services (GuardDuty, Security Command Center, Defender for Cloud) augment on-premises SIEM with cloud-specific behavioral detection
- Zero-trust remote access via Tailscale — cloud nodes expose no management ports to the public internet
- Unified DNS via on-premises Unbound resolvers provides consistent name resolution, ad/malware blocking, and DNSSEC validation for all cloud nodes
- PatchMon, Checkmk, and Ansible extend on-premises patch management, monitoring, and configuration enforcement to cloud workloads
- Routing isolation per cloud provider prevents unintended lateral movement between cloud environments

Deployment Rationale

Enterprise environments increasingly operate hybrid architectures where workloads span on-premises and multiple cloud providers. This deployment demonstrates proficiency with cloud-native security services, hybrid network design, and cross-environment monitoring. Each cloud provider is configured to mirror production patterns: IAM least-privilege, VPC/VNet segmentation, native logging, and endpoint agents for unified visibility. The on-premises management toolchain extended to cloud nodes demonstrates a consistent, tool-agnostic operations model applicable across infrastructure boundaries.

Architecture Principles Alignment

Defense in Depth

- Multiple cloud providers create independent security domains with distinct native controls
- Wazuh EDR coverage spans on-premises and cloud nodes for unified endpoint telemetry
- Cloud-native threat detection (GuardDuty, SCC, Defender for Cloud) operates in parallel with on-premises SIEM correlation

Secure by Design

- Cloud nodes deployed with least-privilege IAM/service account roles; no embedded credentials
- All management access routed through Tailscale — no public inbound ports on management interfaces
- Native cloud audit logging (CloudTrail, GCP Audit Logs, Azure Monitor) enabled by default
- On-premises Unbound DNS provides malware domain filtering for all cloud node outbound resolution

Zero Trust

- Tailscale ACLs enforce explicit allow rules between cloud and on-premises nodes
- Cloud firewall rules restrict inbound to Tailnet CGNAT space (100.64.0.0/10) and homelab subnets
- No implicit trust between cloud providers or between cloud and on-premises segments

Tailscale Mesh VPN — Cross-Cloud Fabric

Deployment Overview

Tailscale provides the encrypted WireGuard-based overlay connecting all cloud nodes, on-premises infrastructure, and remote endpoints into a single private network. Each node is enrolled in the tailnet (tailf07c05.ts.net) and receives a stable CGNAT address (100.64.0.0/10). pfSense nodes advertise on-premises subnets into the tailnet; cloud VMs advertise their respective VPC/VNet CIDRs. This enables any tailnet member to reach cloud and on-premises subnets without requiring per-host agent installation on every workload.

Security Impact

- Eliminates public-internet exposure of management interfaces across all cloud platforms
- WireGuard encryption (ChaCha20-Poly1305) secures all inter-node traffic by default
- ACL-based policy enforces least-privilege connectivity — no implicit mesh between all nodes
- Exit node capability routes homelab egress through cloud endpoints for geo-validation and privacy testing
- Device-bound authentication prevents credential-only access; Tailscale identity is tied to machine keys
- On-premises Unbound DNS resolvers accessible to cloud nodes via Tailscale for consistent internal name resolution and filtering

Node Inventory

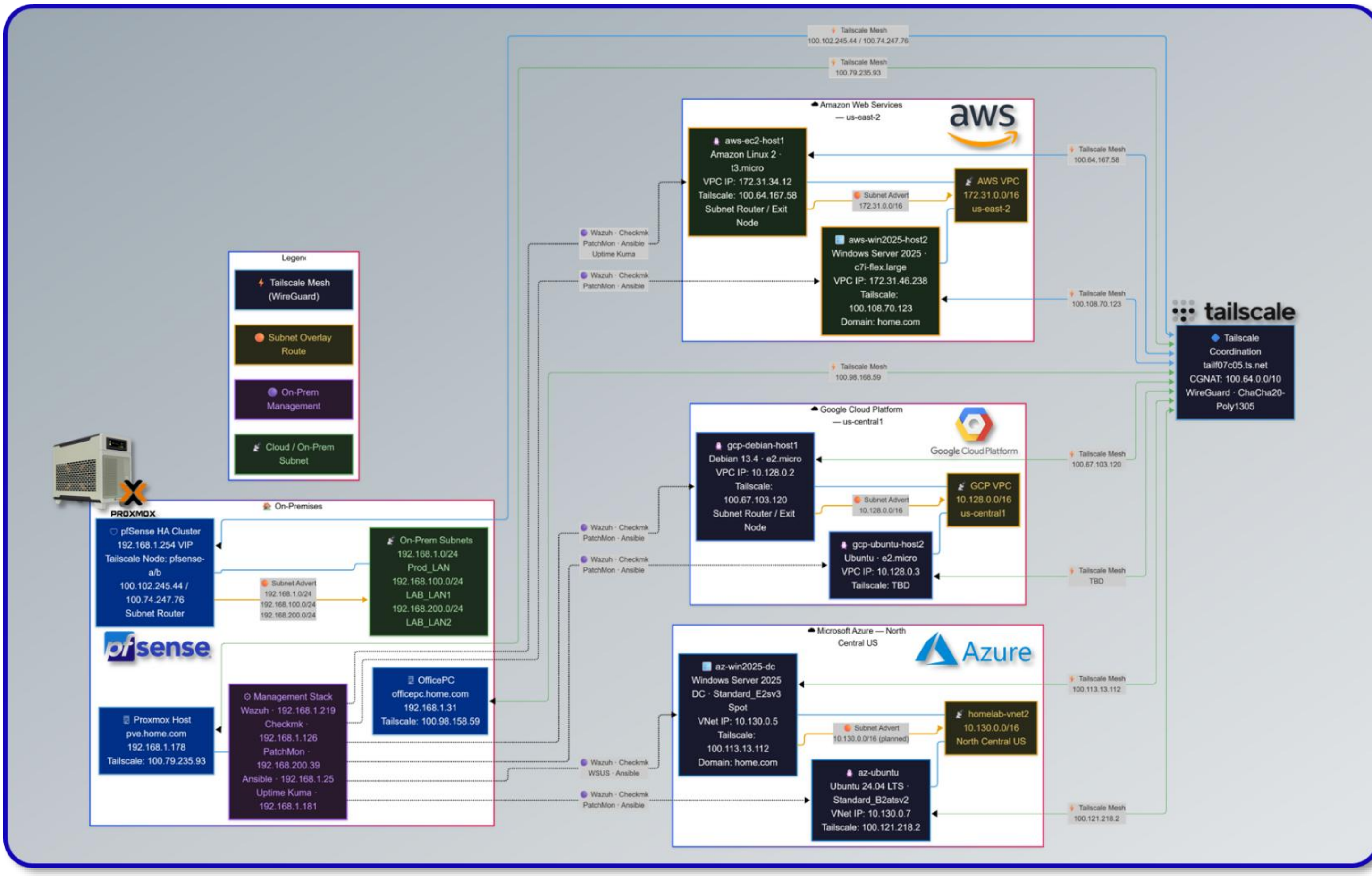
Hostname	Platform	Local IP	Tailscale IP	Role
officepc.home.com	Windows 11 (On-Prem)	192.168.1.31	100.98.158.59	Endpoint
pve.home.com	Proxmox Host (On-Prem)	192.168.1.178	100.79.235.93	Endpoint
pfsense-a (fw.home.com)	pfSense FreeBSD 15	192.168.100.2	100.102.245.44	Subnet Router / Exit Node
pfsense-b (fw.home.com)	pfSense FreeBSD 15	192.168.100.3	100.74.247.76	Subnet Router / Exit Node
aws-ec2-host1	Amazon Linux (AWS EC2)	172.31.34.12	100.64.167.58	Subnet Router / Exit Node
aws-win2025-host2	Windows Server 2025 (AWS)	172.31.46.238	100.108.70.123	Endpoint
gcp-debian-host1	Debian 13.4 (GCP e2.micro)	10.128.0.2	100.67.103.120	Subnet Router / Exit Node
az-ubuntu	Ubuntu 24.04 LTS (Azure B2ats v2)	10.130.0.7	100.96.110.40	Subnet Router / Exit Node
az-win2025-dc	Windows Server 2025 Datacenter (Azure Standard E2s v3)	10.130.0.5	100.113.13.112	Endpoint

Subnet Advertisement

Node	Advertised Subnets	Exit Node
pfsense-a / pfsense-b	192.168.1.0/24, 192.168.100.0/24, 192.168.200.0/24	Yes
aws-ec2-host1	172.31.0.0/16	Yes

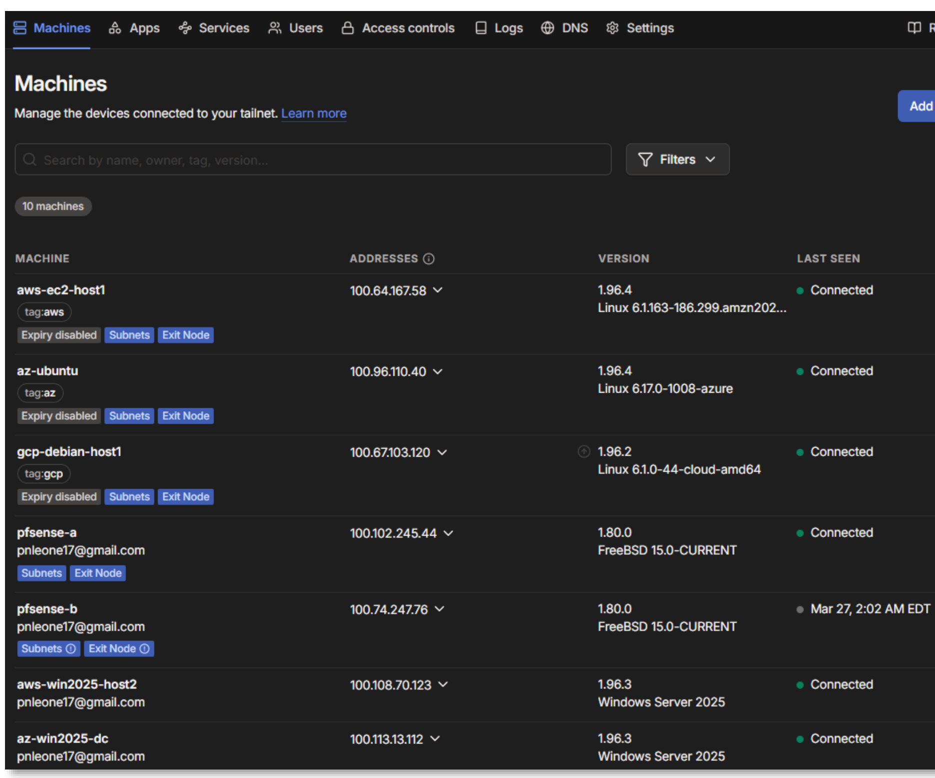
gcp-debian-host1	10.128.0.0/16	Yes
az-ubuntu	10.130.0.0/16	Yes

Routing Overview – Tailscale fabric/underlay and private subnet overlay



DNS via On-Premises Unbound

Cloud nodes resolve internal hostnames and benefit from malware/ad-blocking through the on-premises Unbound resolvers, accessible via the Tailscale tunnel. DNS queries from cloud nodes route over Tailscale to Unbound-01 (192.168.1.153) and Unbound-02 (192.168.1.154) as primary and secondary resolvers. Windows Server nodes that are domain-joined to home.com use the on-premises Active Directory DNS servers (192.168.1.152 / 192.168.1.142) for AD-specific resolution.



Resolver	IP	Role	Access Path
Unbound-01	192.168.1.153	Primary recursive resolver + ad/malware blocking	Via Tailscale tunnel
Unbound-02	192.168.1.154	Secondary recursive resolver (independent cache)	Via Tailscale tunnel
Technitium DNS01	192.168.1.150	Authoritative for home.com (forwarded from Unbound)	Via Tailscale tunnel
Technitium DNS02	192.168.1.151	Secondary authoritative (zone replica)	Via Tailscale tunnel
DC01 / DC02 (AD DNS)	192.168.1.152 / 192.168.1.142	Domain DNS — used by Windows domain-joined cloud nodes	Via Tailscale tunnel

ACL Policy

Tailscale ACLs are maintained in the admin console and enforce explicit allow rules. Default policy denies all traffic not explicitly permitted.

- Homelab endpoints reach all subnet-router-advertised cloud CIDRs
- Cloud nodes reach on-premises Wazuh Manager (192.168.1.219) on TCP 1514/1515
- Cloud nodes reach Unbound DNS resolvers (192.168.1.153/154) on UDP/TCP 53
- Domain-joined Windows cloud nodes reach AD DNS (192.168.1.152/142) and domain controllers on required AD ports
- Cloud nodes reach PatchMon backend, Checkmk server, and Ansible controller via homelab subnets
- SSH access (TCP 22) and RDP (TCP 3389) permitted from homelab subnets to cloud nodes only
- ICMP permitted within tailnet for diagnostics

Security Controls

Control	Implementation	Purpose
Encryption	WireGuard — ChaCha20-Poly1305	All tunnel traffic encrypted in transit
Authentication	Machine keys + Tailscale identity	Device-bound; no shared secrets
ACL Enforcement	Tailscale admin console policy	Least-privilege inter-node connectivity
Exit Node	pfsense-a/b, aws-ec2-host1, gcp-debian-host1	Controlled egress routing
MagicDNS	tailf07c05.ts.net	Stable human-readable hostnames across tailnet
Internal DNS	On-premises Unbound / AD DNS via Tailscale	Consistent resolution and malware blocking
Key Rotation	Automatic (Tailscale managed)	Prevents stale key material

Amazon Web Services (AWS)

Deployment Overview

The AWS deployment consists of two EC2 instances: a t3.micro running Amazon Linux 2 that serves as the Tailscale subnet router for the VPC (172.31.0.0/16), and a c7i-flex.large running Windows Server 2025 Datacenter. Both instances are enrolled in the Tailscale tailnet and integrated with the on-premises management stack (Wazuh, PatchMon, Checkmk, Ansible, Uptime Kuma). The Windows Server 2025 instance is joined to the home.com Active Directory domain via Tailscale, extending enterprise identity and Group Policy to the cloud workload. AWS native services; GuardDuty, Security Hub, CloudTrail, IAM, and CloudWatch. Provide cloud-layer threat detection and audit logging that complements the on-premises SOC platform.

On-premises management integration for both instances is covered in the unified On-Premises Management Integration section. Each instance is enrolled in Wazuh, PatchMon, Checkmk, and Ansible using the same toolchain and playbooks applied to on-premises hosts.

Security Impact

- Wazuh agent provides endpoint telemetry from both instances forwarded to the central on-premises SIEM
- Windows Server 2025 joined to home.com AD — Group Policy, WSUS, and certificate auto-enrollment apply across the VPN
- PatchMon daily checks ensure package currency across on-premises and cloud nodes in a unified dashboard
- Checkmk monitors EC2 host health (CPU, memory, disk, services) alongside all lab infrastructure
- GuardDuty delivers ML-based threat detection across VPC flow logs, CloudTrail events, and DNS query logs
- Security Hub aggregates GuardDuty, Inspector, and IAM Access Analyzer findings into a compliance dashboard
- CloudTrail captures all API activity with tamper-resistant log storage in S3
- Security groups restrict all inbound management to Tailnet CGNAT and homelab subnets

Deployment Rationale

AWS represents the dominant enterprise cloud platform. This deployment demonstrates proficiency with EC2 lifecycle management, VPC networking, IAM policy design, and cloud-native security services. The dual-instance topology, one Linux subnet router and one Windows domain member, mirrors common hybrid enterprise patterns where cloud workloads participate in on-premises identity and management infrastructure. GuardDuty and Security Hub provide continuous compliance visibility applicable to CIS AWS Foundations and SOC 2 requirements.

Architecture Principles Alignment

Defense in Depth

- VPC security groups enforce network-layer segmentation
- GuardDuty behavioral detection operates independent of signature-based rules
- CloudTrail + CloudWatch Logs provide a tamper-resistant audit trail at the cloud control plane

Secure by Design

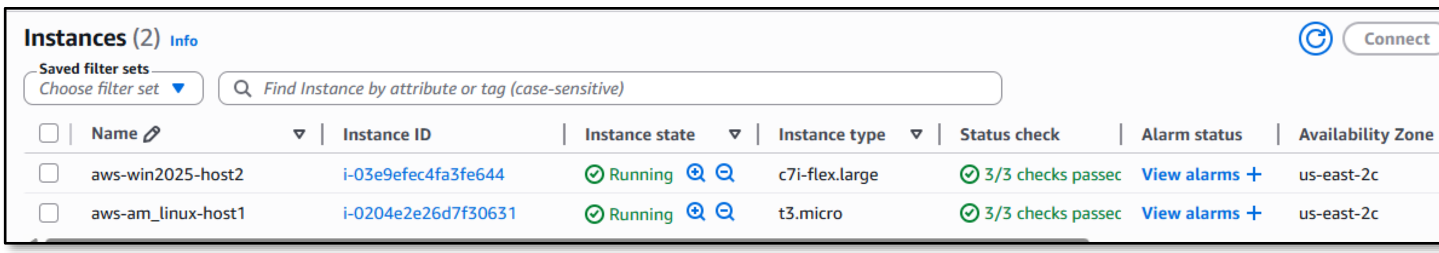
- EC2 instance profiles provide IAM roles without embedded credentials
- Source/Destination check disabled only on the Tailscale Linux node to enable subnet routing — no broad bypass
- S3 bucket for CloudTrail logs configured with server-side encryption
- Windows Server 2025 credential management via domain accounts and WSUS-managed patches

Zero Trust

- No inbound internet-facing management ports, all access via Tailscale
- IAM policies scoped to minimum required actions; no wildcard permissions
- VPC flow logs capture all traffic for forensic analysis

Infrastructure Configuration

EC2 Instances

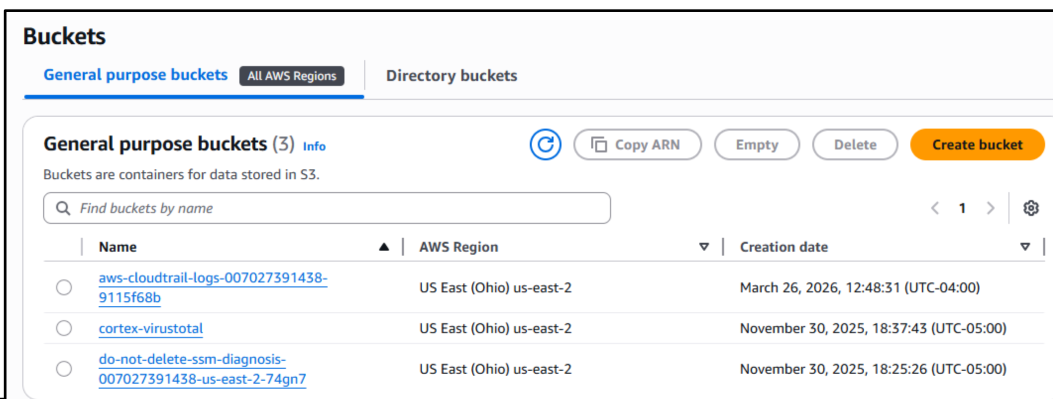


Instances (2) Info

Saved filter sets: Choose filter set — Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
aws-win2025-host2	i-03e9efec4fa3fe644	Running	c7i-flex.large	3/3 checks passec	View alarms +	us-east-2c
aws-am_linux-host1	i-0204e2e26d7f30631	Running	t3.micro	3/3 checks passec	View alarms +	us-east-2c

S3 Buckets



Buckets

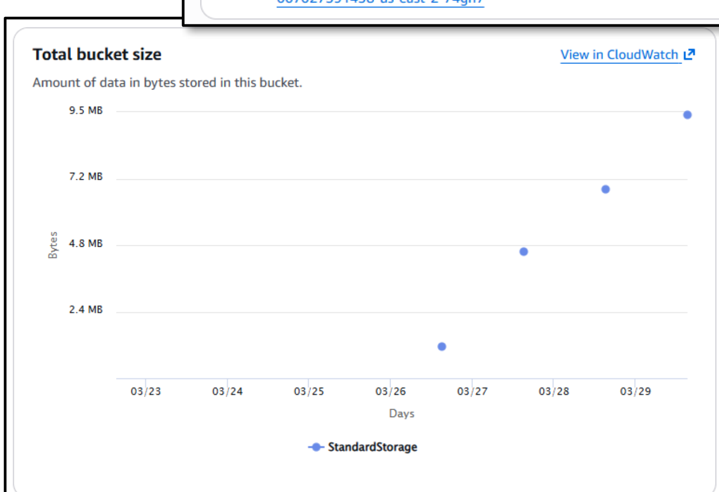
General purpose buckets (All AWS Regions) | Directory buckets

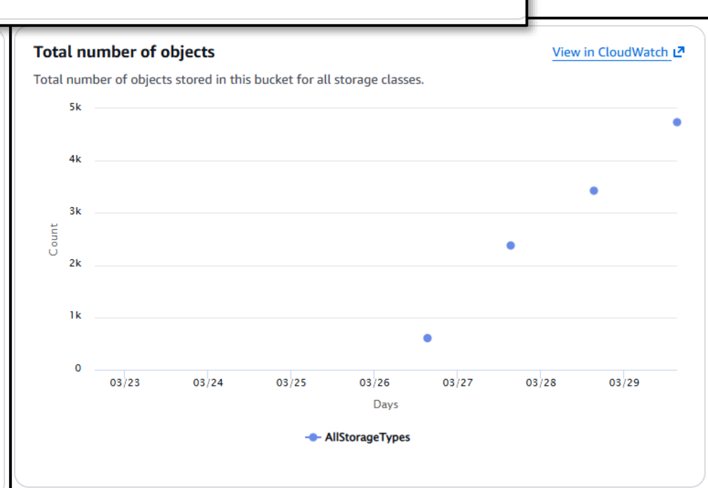
General purpose buckets (3) Info — Copy ARN | Empty | Delete | Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
aws-cloudtrail-logs-007027391438-9115f68b	US East (Ohio) us-east-2	March 26, 2026, 12:48:31 (UTC-04:00)
cortex-virustotal	US East (Ohio) us-east-2	November 30, 2025, 18:37:43 (UTC-05:00)
do-not-delete-ssm-diagnosis-007027391438-us-east-2-74gn7	US East (Ohio) us-east-2	November 30, 2025, 18:25:26 (UTC-05:00)





Amazon Linux 2 — host1 (Subnet Router)

Attribute	Value
Instance ID	i-0204e2e26d7f30631
Hostname	aws-ec2-host1
Instance Type	t3.micro
AMI	Amazon Linux 2 (us-east-2)
Availability Zone	us-east-2c
Private IP	172.31.34.12
Public IP	3.16.91.166 (not used for management — Tailscale only)
Tailscale IP	100.64.167.58
Tailscale Hostname	aws-ec2-host1.tailf07c05.ts.net
Tailscale Role	Subnet Router / Exit Node (172.31.0.0/16)
Volume	8 GiB gp3 (vol-0004b818368cd88bd)
Key Pair	OfficePC (ed25519)
Source/Dest Check	Disabled (required for Tailscale subnet routing)
DNS Resolvers	192.168.1.153, 192.168.1.154 (on-premises Unbound via Tailscale)

Windows Server 2025 Datacenter — host2

Attribute	Value
Instance ID	i-03e9efec4fa3fe644
Hostname	aws-win2025-host2
Instance Type	c7i-flex.large
AMI	Windows Server 2025 Datacenter (us-east-2)
Availability Zone	us-east-2c
Private IP	172.31.34.137
Tailscale IP	100.108.70.123
Tailscale Hostname	aws-win2025-host2.tailf07c05.ts.net
Domain	home.com (domain-joined via Tailscale)
Monitoring	Enabled (CloudWatch)
Security Groups	tailscale-access, lab-services
Key Pair	paul-key2
Source/Dest Check	Disabled
DNS Resolvers	192.168.1.152, 192.168.1.142 (on-premises AD DNS via Tailscale — required for domain join)

VPC Configuration

Attribute	Value
VPC ID	vpc-07db6afa9c6097fe4
CIDR	172.31.0.0/16
Default VPC	Yes
Region	us-east-2 (Ohio)
Internet Gateway	igw-0d9a301fe45791b9b
Route Table	rtb-00a8870decfc6d646 (main)

Routing

Layer	Node / Scope	Destination	Next Hop / Gateway
Tailnet	pfSense-a/b	172.31.0.0/16	Advertised via aws-ec2-host1

Tailnet	aws-ec2-host1	192.168.1.0/24, 192.168.100.0/24, 192.168.200.0/24	Advertised via pfSense Nodes
Home LAN	Office PC (192.168.1.31)	172.31.0.0/16	192.168.1.253 (pfSense)
AWS VPC	VPC Route Table	192.168.0.0/16	ENI of EC2 Tailscale Host
pfSense	Outbound NAT — Tailscale IF	192.168.1.0/24	Default (bypasses PIA VPN)
DNS (Linux)	aws-ec2-host1	192.168.1.153, 192.168.1.154	Via Tailscale tunnel to on-prem Unbound
DNS (Windows)	aws-win2025-host2	192.168.1.152, 192.168.1.142	Via Tailscale tunnel to on-prem AD DNS

Security Groups

lab-services — Inbound Rules

Protocol	Port	Source	Description
TCP	22	192.168.0.0/16	SSH — homelab subnets only (Linux)
TCP	3389	192.168.0.0/16	RDP — homelab subnets only (Windows)
ICMP	All	192.168.0.0/16	Internal diagnostics
TCP	80	192.168.1.181/32	Uptime Kuma — HTTP monitor
UDP	33434-33600	192.168.0.0/16	Internal traceroute

lab-services — Outbound Rules

Protocol	Port(s)	Destination	Description
TCP	1514-1515, 55000	192.168.1.219/32	Wazuh agent ports
TCP/UDP	53	192.168.1.153/32	Unbound-01 DNS
TCP/UDP	53	192.168.1.154/32	Unbound-02 DNS
All TCP	*	192.168.1.126/32	Checkmk agent
TCP	3000-3001	192.168.200.0/24	PatchMon ports
ICMP	*	192.168.0.0/16	Ping testing
UDP	33434-33600	192.168.0.0/16	Traceroute
All Traffic	AD Ports	192.168.1.152/32, 192.168.1.142/32	AD Domain Controller — Kerberos, LDAP, DNS, GP, SYSVOL

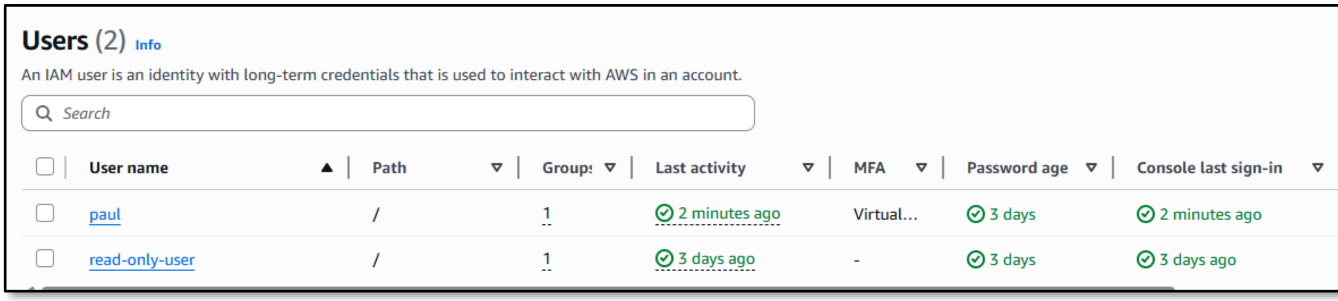
tailscale-access — Inbound / Outbound

Protocol	Port	Source / Destination	Description
All	All	100.64.0.0/10	Bidirectional Tailnet communication

IAM Configuration

Resource	Count	Notes
User Groups	2	Lab admin group, read-only group
Users	2	Admin and read-only users — no long-lived access keys on EC2

Roles	24	Service-linked + custom instance profiles
Policies	1	Custom least-privilege policy
Identity Providers	0	Not configured



Native Security Services

GuardDuty

Continuous threat detection analyzing VPC flow logs, CloudTrail management events, and Route 53 DNS query logs. ML-based anomaly detection identifies reconnaissance, credential compromise, and C2 communication without rule maintenance.

Data Source	Detection Focus
VPC Flow Logs	Port scanning, unusual outbound connections, lateral movement
CloudTrail Logs	Unauthorized API calls, credential misuse, privilege escalation
DNS Query Logs	DGA domains, DNS exfiltration, C2 callbacks

Security Hub

Aggregates findings from GuardDuty, Inspector, and IAM Access Analyzer. Benchmarks include CIS AWS Foundations and AWS Foundational Security Best Practices. Findings export to CloudWatch Events for automated response via Lambda.

CloudTrail

Management and API audit logging enabled for all events. Logs delivered to S3 with server-side encryption. CloudWatch Logs integration enables real-time alerting on high-risk API calls including IAM policy changes, security group modifications, and root account usage.

User name	Event time	Event source	Event name
Paul	2026-03-28T13:25:18Z	ec2.amazonaws.com	StartInstances
Paul	2026-03-28T13:24:49Z	signin.amazonaws.com	ConsoleLogin
Paul	2026-03-28T13:24:32Z	signin.amazonaws.com	CheckMfa
read-only-user	2026-03-28T13:22:08Z	ec2.amazonaws.com	StartInstances
read-only-user	2026-03-28T13:21:13Z	signin.amazonaws.com	ConsoleLogin
Paul	2026-03-28T13:13:28Z	ec2.amazonaws.com	StopInstances
Paul	2026-03-28T13:13:20Z	ec2.amazonaws.com	StopInstances
Paul	2026-03-28T11:43:29Z	logs.amazonaws.com	DeleteLogStream
Paul	2026-03-27T13:03:09Z	ec2.amazonaws.com	ModifySecurityGroupRules
Paul	2026-03-27T13:01:17Z	ec2.amazonaws.com	AuthorizeSecurityGroupIngress
Paul	2026-03-27T13:01:17Z	ec2.amazonaws.com	ModifySecurityGroupRules
Paul	2026-03-26T16:51:51Z	cloudtrail.amazonaws.com	UpdateTrail
CloudTrail	2026-03-26T16:51:51Z	logs.amazonaws.com	CreateLogStream
CloudTrail	2026-03-26T16:51:49Z	logs.amazonaws.com	CreateLogStream
Paul	2026-03-26T16:51:49Z	cloudtrail.amazonaws.com	UpdateTrail
Paul	2026-03-26T16:51:47Z	cloudtrail.amazonaws.com	UpdateTrail
CloudTrail	2026-03-26T16:51:47Z	logs.amazonaws.com	CreateLogStream
Paul	2026-03-26T16:51:45Z	cloudtrail.amazonaws.com	UpdateTrail
CloudTrail	2026-03-26T16:51:45Z	logs.amazonaws.com	CreateLogStream
Paul	2026-03-26T16:51:42Z	cloudtrail.amazonaws.com	UpdateTrail
CloudTrail	2026-03-26T16:51:42Z	logs.amazonaws.com	CreateLogStream
Paul	2026-03-26T16:51:40Z	cloudtrail.amazonaws.com	UpdateTrail
Paul	2026-03-26T16:48:31Z	cloudtrail.amazonaws.com	PutEventSelectors
Paul	2026-03-26T16:48:31Z	cloudtrail.amazonaws.com	CreateTrail
Paul	2026-03-26T16:48:31Z	cloudtrail.amazonaws.com	StartLogging
Paul	2026-03-26T16:48:31Z	s3.amazonaws.com	PutBucketEncryption
Paul	2026-03-26T16:48:30Z	s3.amazonaws.com	PutBucketPolicy
Paul	2026-03-26T16:48:30Z	s3.amazonaws.com	CreateBucket
Paul	2026-03-26T14:22:52Z	ec2.amazonaws.com	AuthorizeSecurityGroupEgress
Paul	2026-03-26T14:00:24Z	ec2.amazonaws.com	AuthorizeSecurityGroupEgress
Paul	2026-03-26T13:51:13Z	ec2.amazonaws.com	CreateTags
Paul	2026-03-26T13:49:28Z	ec2.amazonaws.com	AuthorizeSecurityGroupEgress
Paul	2026-03-26T13:48:58Z	ec2.amazonaws.com	AuthorizeSecurityGroupIngress
vpc-flow-logging	2026-03-26T13:32:45Z	logs.amazonaws.com	CreateLogStream

CloudWatch

Infrastructure metrics collected from both EC2 instances via CloudWatch Agent. Custom metrics and log groups forward application and system logs. Alarms trigger SNS notifications and can initiate Lambda-based automated response.

VPC Flow Logs (flow log ID: fl-03179f74e54bf1aa4) are delivered to CloudWatch Logs group RDSOSMetrics, capturing all accepted and rejected traffic with ENI, source/destination addresses, ports, and action. This provides granular visibility into traffic between AWS instances and on-premises subnets via the Tailscale tunnel.

Monitoring and Alerting

Source	Alert Type	Destination
GuardDuty findings	Threat detection (high/medium severity)	Security Hub + CloudWatch Events
CloudTrail	High-risk API activity	CloudWatch Alarms + SNS
CloudWatch Agent	EC2 resource thresholds	CloudWatch Alarms
Wazuh Agent	Endpoint security events	On-premises Wazuh Manager + Splunk SIEM
PatchMon Agent	Outdated packages / CVE correlation	On-premises PatchMon dashboard + Discord
Checkmk Agent	Host health (CPU, disk, services)	On-premises Checkmk + Discord
Uptime Kuma	Service availability checks	Discord webhook (#uptime-kuma)

Google Cloud Platform (GCP)

Deployment Overview

The GCP deployment consists of two e2.micro instances in the us-central1-c zone. The primary node (gcp-debian-host1) runs Debian 13.4 and serves as the Tailscale subnet router for the GCP VPC (10.128.0.0/16), advertising the GCP network into the tailnet as an exit node. A secondary Ubuntu instance (gcp-ubuntu-host2) operates as an additional workload endpoint. Both instances are enrolled in Tailscale and integrated with the on-premises management stack. GCP native services — Security Command Center, Cloud Logging, Cloud Monitoring, IAM, Cloud Armor, and VPC firewall policies — provide cloud-native security visibility that complements the on-premises SOC platform.

On-premises management integration for both instances is covered in the unified On-Premises Management Integration section.

Security Impact

- Tailscale subnet router on gcp-debian-host1 advertises 10.128.0.0/16 into the tailnet, enabling homelab access to all GCP workloads without public internet exposure
- Wazuh agents on both GCP instances provide endpoint telemetry forwarded to the central on-premises Wazuh Manager
- Security Command Center delivers asset inventory, vulnerability findings, and threat detection across the GCP project
- Cloud Armor policies provide WAF protection (SQLi/XSS blocking), DDoS rate limiting, and geo-based threat intelligence blocking across three policy tiers
- VPC network firewall policy (homelab) enforces Google Cloud Threat Intelligence blocking for TOR exit nodes, known malicious IPs, and sanctioned countries at the network perimeter
- Cloud Logging captures Admin Activity and Data Access audit logs for all GCP API calls
- On-premises Unbound DNS resolvers provide consistent name resolution and malware domain filtering via Tailscale

Deployment Rationale

GCP provides unique cloud security capabilities including Security Command Center for CSPM, Cloud Armor for WAF/DDoS protection with Google Threat Intelligence integration, and a strong IAM model with service accounts and Workload Identity. This deployment demonstrates proficiency with GCP-specific security tooling, multi-cloud subnet routing via Tailscale, and the extension of the on-premises management toolchain to a second cloud provider. The layered Cloud Armor and VPC firewall policy architecture mirrors enterprise-grade GCP deployments where threat intelligence, rate limiting, and WAF rules operate at the edge before traffic reaches compute resources.

Architecture Principles Alignment

Defense in Depth

- VPC network firewall policy (homelab) provides the innermost perimeter — Google Threat Intelligence blocks TOR exit nodes, known malicious IPs, and sanctioned countries
- Cloud Armor adds edge-layer WAF and DDoS protection with three policy tiers (tailscale, vm-ddos, application-level)

- Security Command Center provides independent asset and vulnerability visibility above network controls
- Cloud Logging audit trail captures all control-plane API activity

Secure by Design

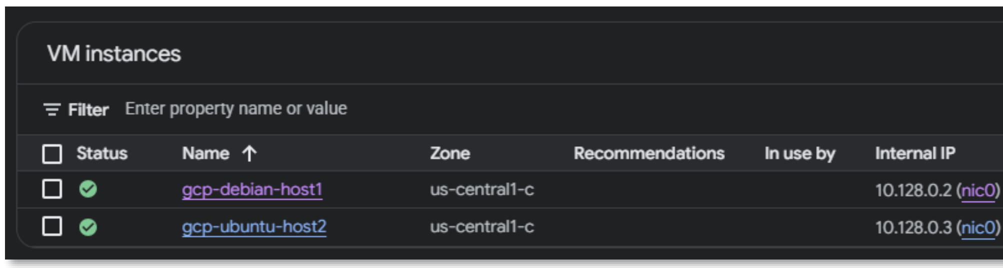
- GCP service accounts with minimal IAM roles; no user credentials on instances
- Cloud Armor application-level policy blocks SQLi and XSS before traffic reaches application backends
- VPC firewall policy restricts management access (SSH) to homelab subnets — public-internet exposure removed

Zero Trust

- Management access exclusively via Tailscale — public IP used only for Tailscale UDP handshake (port 41641)
- IAM service accounts follow least-privilege; no owner or editor roles assigned to instances
- VPC flow logs enabled for forensic traffic analysis

Infrastructure Configuration

VM Instances

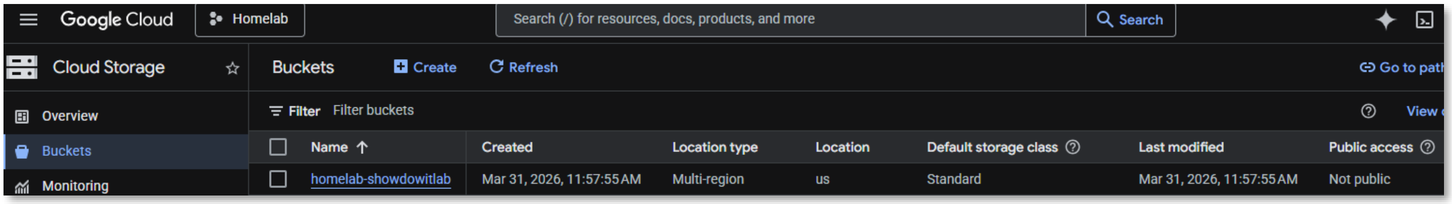


Attribute	gcp-debian-host1	gcp-ubuntu-host2
OS	Debian 13.4	Ubuntu
Machine Type	e2.micro	e2.micro
Zone	us-central1-c	us-central1-c
Internal IP	10.128.0.2 (nic0)	10.128.0.3 (nic0)
External IP	34.29.32.124 (nic0)	108.59.80.1 (nic0)
Tailscale IP	100.67.103.120	TBD
Tailscale Hostname	gcp-debian-host1.tailf07c05.ts.net	gcp-ubuntu-host2.tailf07c05.ts.net
Tailscale Role	Subnet Router / Exit Node	Endpoint
Exit Node Subnet	10.128.0.0/16	N/A
DNS Resolvers	192.168.1.153, 192.168.1.154 (via Tailscale)	192.168.1.153, 192.168.1.154 (via Tailscale)

VPC and Network Configuration

Attribute	Value
VPC Network	default
Project ID	project-9d2957e1-842b-4b35-ad2
Region	us-central1
Zone	us-central1-c
Subnet CIDR	10.128.0.0/16 (auto-mode VPC)
Exit Node Subnet (Tailnet)	10.128.0.0/16
Internet Gateway	Default (GCP managed)
SMTP Port 25	Blocked by GCP project policy

Storage Bucket



Routing

Layer	Node / Scope	Destination	Next Hop / Gateway
Tailnet	pfSense-a/b	10.128.0.0/16	Advertised via gcp-debian-host1
Tailnet	gcp-debian-host1	192.168.1.0/24, 192.168.100.0/24, 192.168.200.0/24	Advertised via pfSense Nodes
GCP VPC	Internal routing	10.128.0.0/9	VPC local (default-allow-internal)
DNS	GCP Instances	192.168.1.153, 192.168.1.154	Via Tailscale tunnel to on-prem Unbound

Cloud Armor Security Policies

Three Cloud Armor policies enforce edge-layer security for GCP workloads, operating at edge, backend, and application tiers. Policies are evaluated in the order listed; lower-priority numbers are evaluated first within each tier.

tailscale — Edge Security Policy

Applied at the global load balancer level. Restricts access to Tailscale network and homelab ranges with a default deny for all other sources.

Priority	Action	Type	Match	Description
80	Allow	IP ranges	100.64.0.0/10	Tailscale network
100	Allow	IP ranges	192.168.0.0/16	Homelab access
2,147,483,647	Deny (403)	IP ranges	* (All)	Default deny — higher priority rules override

vm-ddos — Backend Security Policy

Backend security policy providing DDoS rate limiting for VM-level traffic.

Priority	Action	Type	Match	Description
80	Allow	IP ranges	100.64.0.0/10	Tailscale access
100	Allow	IP ranges	192.168.0.0/16	Homelab access
1,000	Throttle	IP ranges	*	Rate limiting — prevents cost spikes and DDoS
2,147,483,647	Deny	IP ranges	* (All)	Default deny

application-level — Backend Security Policy

Application-layer WAF policy blocking OWASP Top 10 attack patterns. SQLi and XSS rules use GCP's preconfigured expression sets (v33-stable).

Priority	Action	Type	Match / Expression	Description

10	Deny	WAF rule	evaluatePreconfiguredExpr('sqli-v33-stable')	WAF: Block SQL Injection
20	Deny	WAF rule	evaluatePreconfiguredExpr('xss-v33-stable')	WAF: Block XSS
80	Allow	IP ranges	100.64.0.0/10	Tailscale overlay access
100	Allow	IP ranges	192.168.0.0/16	Homelab direct access
1,000	Throttle	IP ranges	*	Rate limit — prevents lab cost spikes
2,147,483,647	Deny (403)	IP ranges	* (All)	Default deny

VPC Network Firewall Policy — homelab

The homelab VPC network firewall policy is attached to the default VPC network (42 subnets, global scope) and contains 17 rules. It replaces and extends basic per-instance firewall rules with centralized policy management, Google Cloud Threat Intelligence integration, and explicit service-level egress rules. Rules are evaluated in priority order — lower numbers first.

Threat Intelligence and Geo-Blocking (Priority 100-130)

Priority	Direction	Source / Destination	Protocols	Action	Description
100	Ingress	Google Threat Intel: TOR exit nodes	All	Deny	Block TOR exit node ingress traffic
110	Ingress	Google Threat Intel: Known malicious IPs	All	Deny	Block known malicious IP ingress
120	Egress	Google Threat Intel: Known malicious IPs	All	Deny	Block egress to known malicious IPs
130	Ingress	Geolocations: CU, IR, KP, SY, XC (Crimea), XD (Donetsk/Luhansk)	All	Deny	Block sanctioned country ingress

Egress — On-Premises Management Services (Priority 1000-2030)

Priority	Direction	Destination	Protocols / Ports	Description
1000	Egress	192.168.1.153/32, 192.168.1.154/32	TCP/UDP 53	DNS — Unbound-01 and Unbound-02
2005	Egress	192.168.1.219/32	TCP 1514, 1515, 55000	Wazuh agent — manager ports
2010	Egress	192.168.1.126/32	All	Checkmk agent egress
2020	Egress	192.168.200.0/24	TCP 3000-3001	PatchMon backend

Ingress — Homelab Access (Priority 1010-2030)

Priority	Direction	Source	Protocols / Ports	Description
1010	Ingress	192.168.0.0/16	TCP 3389	RDP from homelab
1020	Ingress	192.168.0.0/16, 172.31.0.0/16	ICMP	Diagnostics — homelab + AWS
1030	Ingress	192.168.0.0/16	UDP 33434-33600	Traceroute from homelab
1040	Ingress	192.168.0.0/16	TCP 22	SSH from homelab only
2030	Ingress	192.168.1.181/32	TCP 80, 443	Uptime Kuma health checks

Default Rules (Priority 2147483644-2147483647)

GCP-managed default rules pass all remaining IPv4 and IPv6 traffic to the next evaluation stage. These do not explicitly allow or deny; Cloud Armor policies apply before this stage.

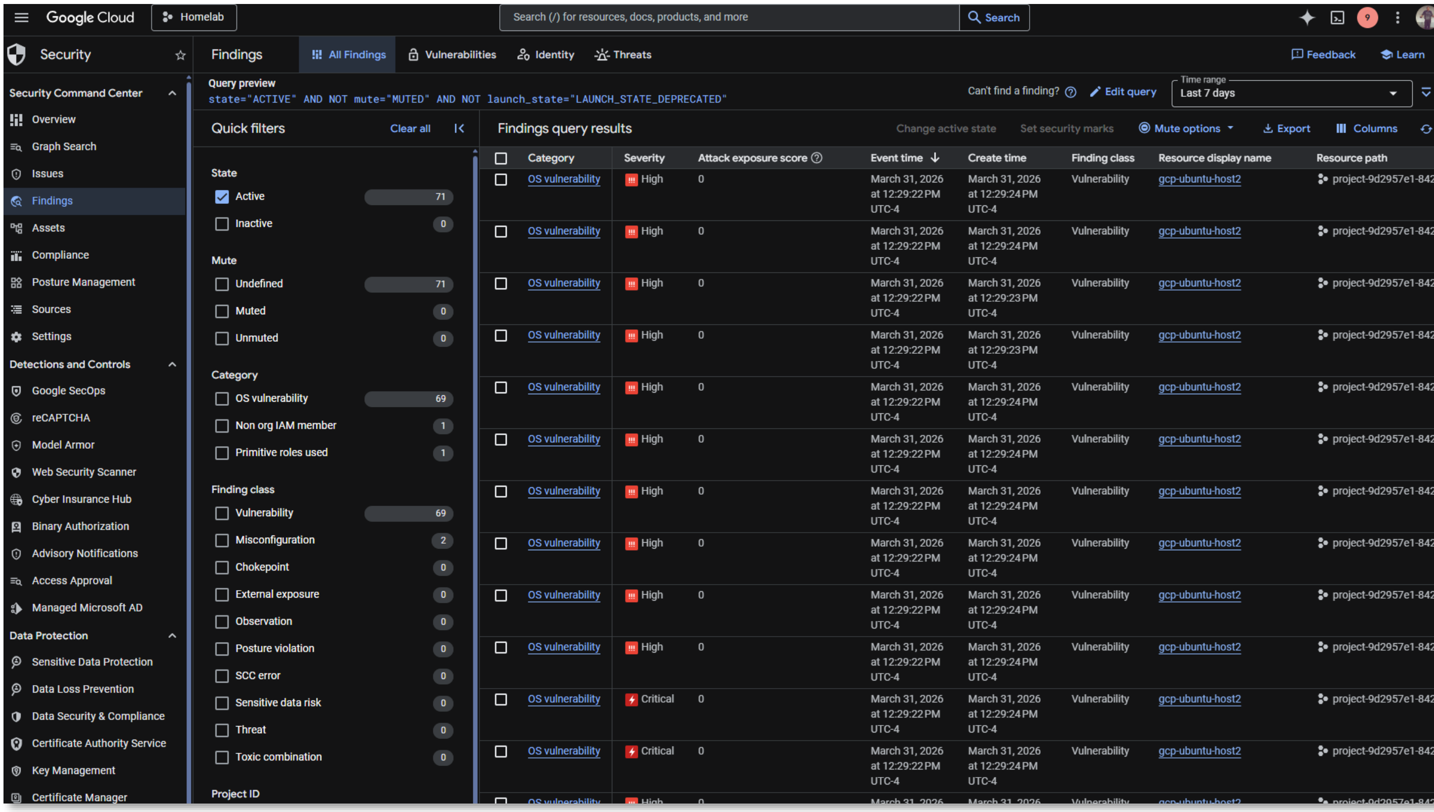
Priority	Direction	Match	Action	Description
2147483644	Egress	IPv6 ::/0	Go to next	Default IPv6 egress pass
2147483645	Ingress	IPv6 ::/0	Go to next	Default IPv6 ingress pass
2147483646	Egress	IPv4 0.0.0.0/0	Go to next	Default IPv4 egress pass
2147483647	Ingress	IPv4 0.0.0.0/0	Go to next	Default IPv4 ingress pass

Native Security Services

Security Command Center

SCC provides asset inventory, vulnerability assessment, misconfiguration detection, and threat detection across the GCP project. Standard tier provides continuous asset discovery and security health analytics.

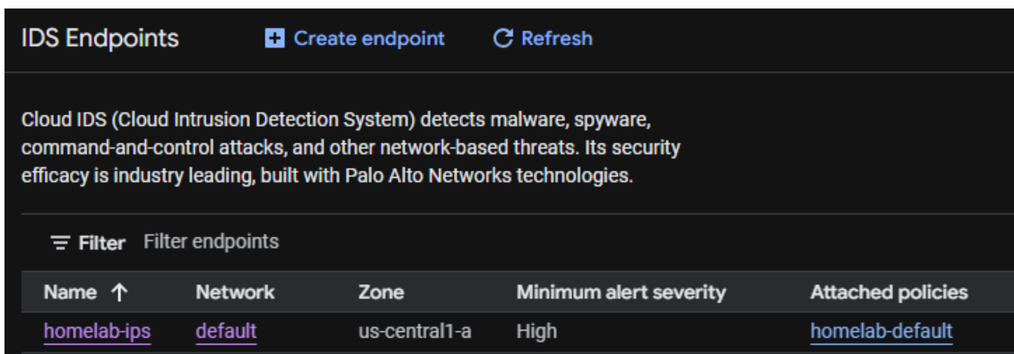
SCC Capability	Function
Security Health Analytics	Detects misconfigurations: open firewall rules, public IPs, weak IAM policies
Asset Inventory	Continuous discovery of all GCP resources across the project
Vulnerability Assessment	CVE scanning for known vulnerabilities on Compute Engine instances
Threat Detection (Premium)	Behavioral anomaly detection for credential misuse and data exfiltration (upgrade path)

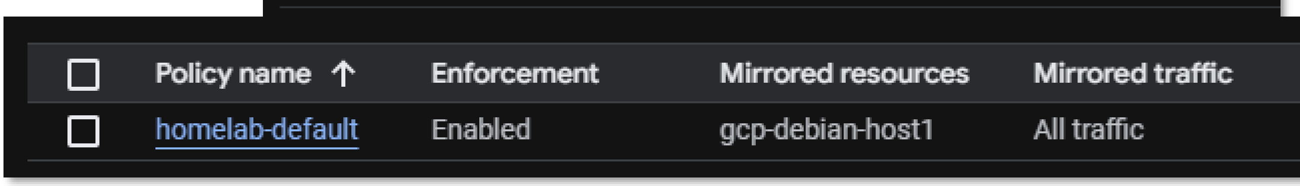


Network Security

IDS Endpoint

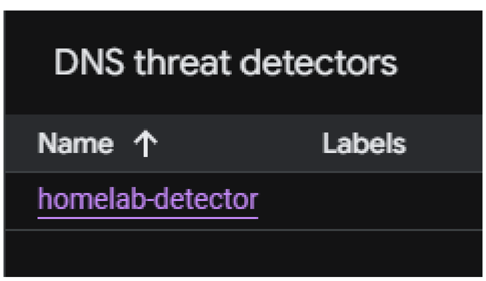
GCP is configured to mirror a single instance, gcp-debian-host1 for all traffic with a minimum lart severity of high.





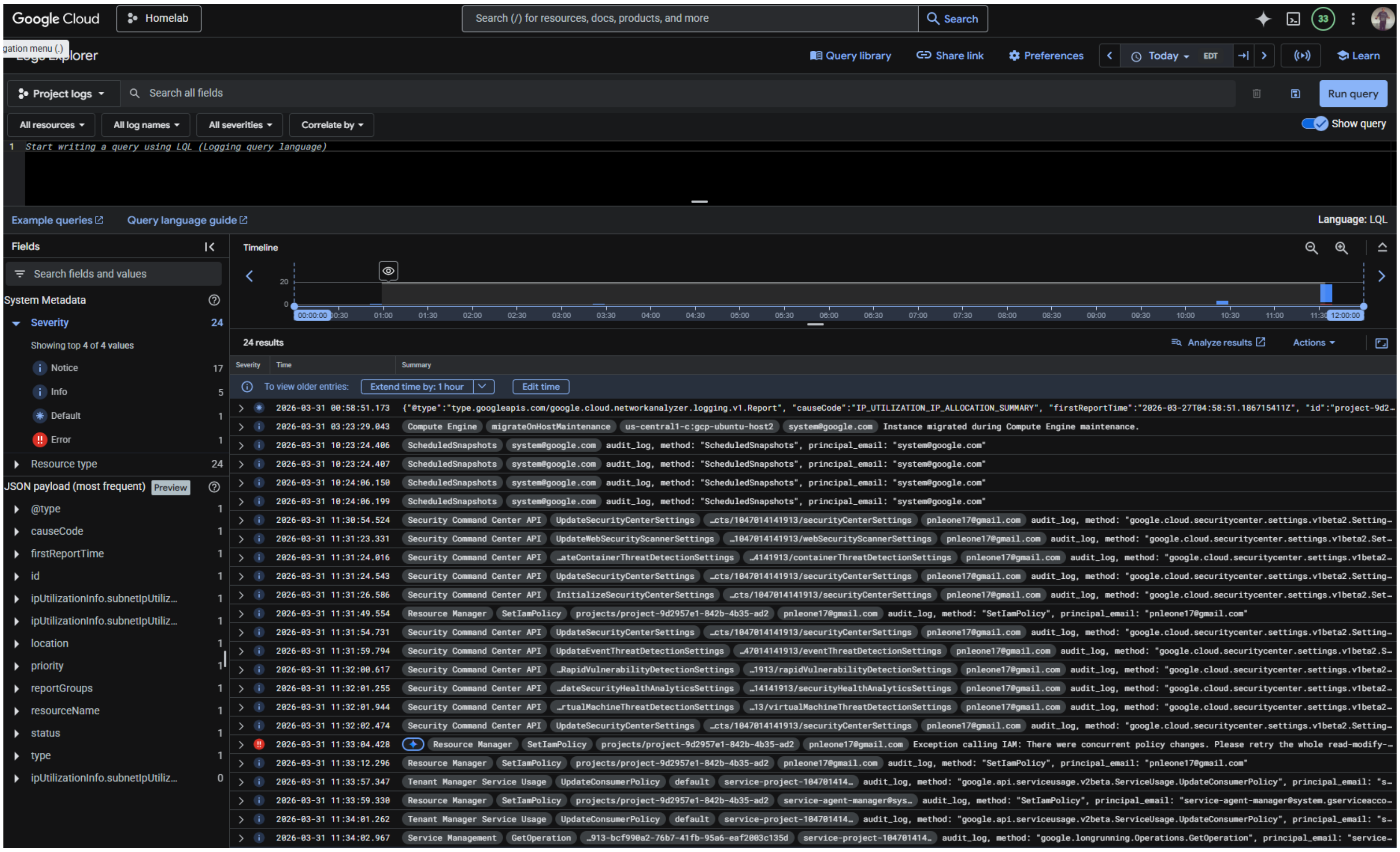
timestamp	ResourceID	resource.type	Dest IP	Dest	Source IP	Source Port	Protocol	packets	location
'2026-03-31T18:26:48.597138745Z	'homelab-ips	ids.googleapis.com/Endpoint	172.217.214.95	443	10.128.0.2	43686	tcp	4	'us-central1-a
'2026-03-31T18:26:48.596846559Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45976	tcp	64	'us-central1-a
'2026-03-31T18:26:48.596730411Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	46032	tcp	48	'us-central1-a
'2026-03-31T18:26:48.596587275Z	'homelab-ips	ids.googleapis.com/Endpoint	172.217.214.95	443	10.128.0.2	43666	tcp	4	'us-central1-a
'2026-03-31T18:26:48.596186144Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45942	tcp	61	'us-central1-a
'2026-03-31T18:26:48.596047425Z	'homelab-ips	ids.googleapis.com/Endpoint	172.217.214.95	443	10.128.0.2	43654	tcp	4	'us-central1-a
'2026-03-31T18:26:48.595740027Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45994	tcp	42	'us-central1-a
'2026-03-31T18:26:48.595327384Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45998	tcp	49	'us-central1-a
'2026-03-31T18:26:48.565893912Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	46044	tcp	39	'us-central1-a
'2026-03-31T18:26:48.565786685Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.36.174	443	10.128.0.2	57292	tcp	25	'us-central1-a
'2026-03-31T18:26:48.565626915Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	46026	tcp	48	'us-central1-a
'2026-03-31T18:26:48.565367833Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36144	tcp	45	'us-central1-a
'2026-03-31T18:26:48.564902796Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	49024	tcp	4	'us-central1-a
'2026-03-31T18:26:46.595109562Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48970	tcp	4	'us-central1-a
'2026-03-31T18:26:44.602999107Z	'homelab-ips	ids.googleapis.com/Endpoint	172.217.214.95	443	10.128.0.2	46566	tcp	4	'us-central1-a
'2026-03-31T18:26:44.602785199Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	49032	tcp	4	'us-central1-a
'2026-03-31T18:26:44.602555997Z	'homelab-ips	ids.googleapis.com/Endpoint	172.217.214.95	443	10.128.0.2	46560	tcp	4	'us-central1-a
'2026-03-31T18:26:44.601228148Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45962	tcp	34	'us-central1-a
'2026-03-31T18:26:44.600945658Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48990	tcp	4	'us-central1-a
'2026-03-31T18:26:44.599189468Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36086	tcp	55	'us-central1-a
'2026-03-31T18:26:44.598229539Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45952	tcp	51	'us-central1-a
'2026-03-31T18:26:44.598006084Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	49010	tcp	4	'us-central1-a
'2026-03-31T18:26:44.597488261Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48992	tcp	4	'us-central1-a
'2026-03-31T18:26:44.596969491Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48952	tcp	4	'us-central1-a
'2026-03-31T18:26:44.596033368Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36098	tcp	40	'us-central1-a
'2026-03-31T18:26:44.566019812Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48980	tcp	4	'us-central1-a
'2026-03-31T18:26:44.565664222Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	45940	tcp	38	'us-central1-a
'2026-03-31T18:26:44.565344915Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48924	tcp	4	'us-central1-a
'2026-03-31T18:26:44.565116778Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36116	tcp	47	'us-central1-a
'2026-03-31T18:26:44.564918344Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48894	tcp	4	'us-central1-a
'2026-03-31T18:26:40.597773525Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48960	tcp	4	'us-central1-a
'2026-03-31T18:26:40.597307489Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48938	tcp	4	'us-central1-a
'2026-03-31T18:26:40.597067186Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36108	tcp	71	'us-central1-a
'2026-03-31T18:26:40.596856334Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36130	tcp	37	'us-central1-a
'2026-03-31T18:26:40.596736193Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36096	tcp	38	'us-central1-a
'2026-03-31T18:26:40.596628267Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36004	tcp	48	'us-central1-a
'2026-03-31T18:26:40.596491329Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36088	tcp	43	'us-central1-a
'2026-03-31T18:26:40.596373698Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48884	tcp	4	'us-central1-a
'2026-03-31T18:26:40.596118986Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	36066	tcp	40	'us-central1-a
'2026-03-31T18:26:40.595968540Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48856	tcp	4	'us-central1-a
'2026-03-31T18:26:40.595645319Z	'homelab-ips	ids.googleapis.com/Endpoint	216.239.34.174	443	10.128.0.2	48914	tcp	4	'us-central1-a

DNS Armor



Cloud Logging

Admin Activity audit logs are enabled by default and cannot be disabled. Data Access logs capture all GCP API calls. Log entries include resource type, method, principal identity, source IP, and request/response metadata. Log-based metrics and alerting policies surface high-risk events such as IAM role modifications and firewall rule changes.



Observability Analytics

Query Recent (3) Saved (0) Save

```
WITH
  flowLogs AS (
  SELECT
    JSON_VALUE(json_payload.connection.src_ip) AS src_ip,
    JSON_VALUE(json_payload.connection.dest_ip) AS dest_ip,
    JSON_VALUE(json_payload.src_instance.vm_name) AS src_instance_name,
    JSON_VALUE(json_payload.dest_instance.vm_name) AS dest_instance_name,
    JSON_VALUE(json_payload.src_instance.project_id) AS src_instance_project_id,
    JSON_VALUE(json_payload.dest_instance.project_id) AS dest_instance_project_id
```

This query will process 6.32 MB when run.

Format Clear

Results (101) Table Chart Both Download Create alert

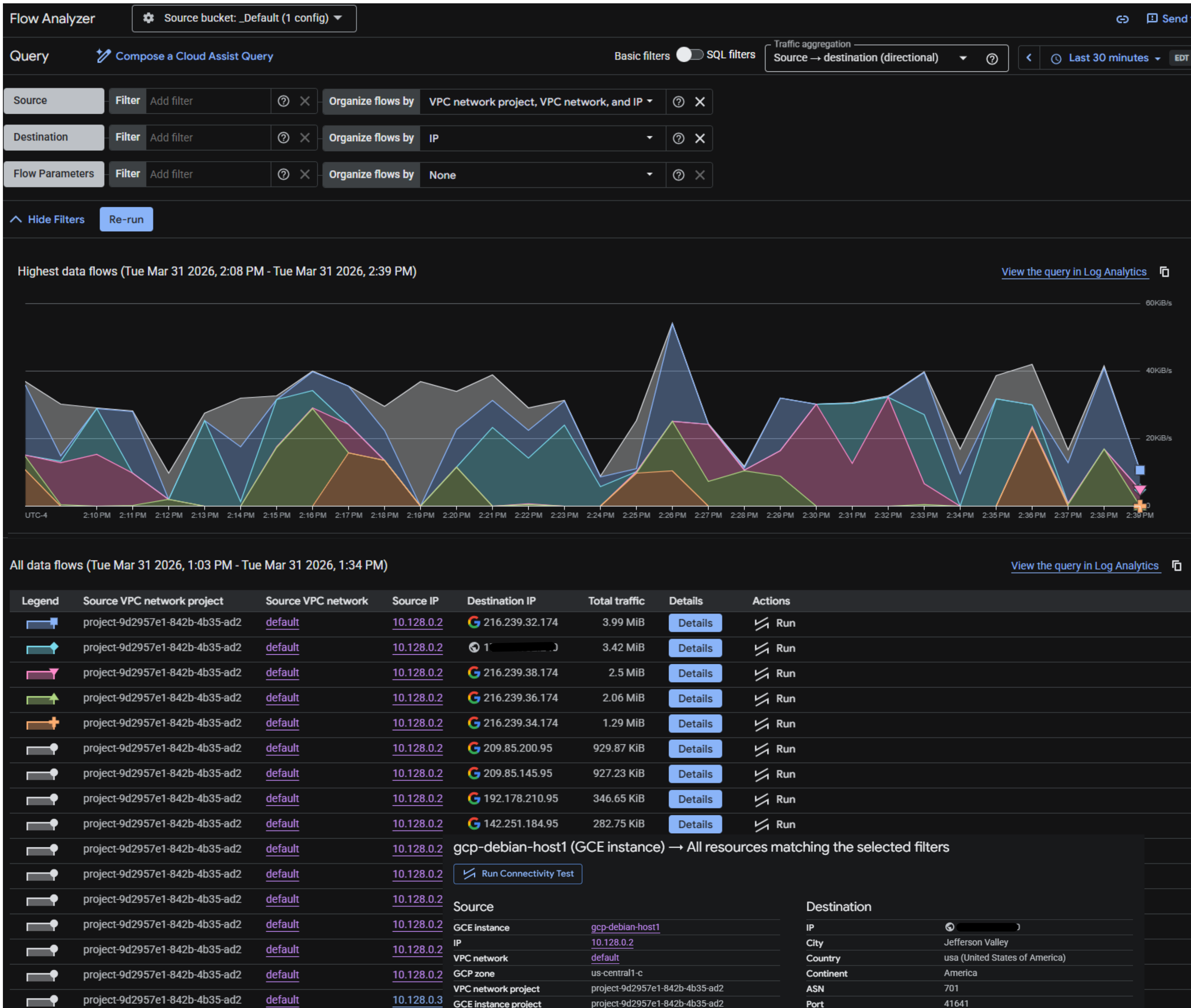
Row	src_vpc_project_id STRING	src_vpc_name STRING	src_ip STRING	dest_ip STRING	src_instance_project_id STRING	src_gcp_zone STRING	src_instance_name STRING
1	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	216.239.32.174	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
2	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	216.239.38.174	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
3	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	216.239.36.174	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
4	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	209.85.200.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
5	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	216.239.34.174	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
6	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	142.250.152.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
7	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	173.68.102.240	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
8	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	64.233.181.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
9	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	142.251.183.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
10	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	173.194.194.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
11	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	142.250.125.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
12	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	64.233.179.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
13	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	192.178.129.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
14	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	74.125.202.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
15	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	74.125.69.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
16	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	173.194.206.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
17	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	172.217.214.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
18	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	108.177.121.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
19	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	74.125.201.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
20	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	74.125.126.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
21	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	74.125.132.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
22	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	173.194.195.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
23	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	199.165.136.101	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
24	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	192.168.2.1	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
25	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	239.255.255.250	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
26	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	209.85.145.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
27	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	192.168.2.2	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
28	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	192.178.209.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
29	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	10.10.0.2	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
30	project-9d2957e1-842b-4b35-ad2	default	10.128.0.3	64.233.179.95	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-ubuntu-host2
31	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	192.168.1.254	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1
32	project-9d2957e1-842b-4b35-ad2	default	10.128.0.2	192.168.200.2	project-9d2957e1-842b-4b35-ad2	us-central1-c	gcp-debian-host1

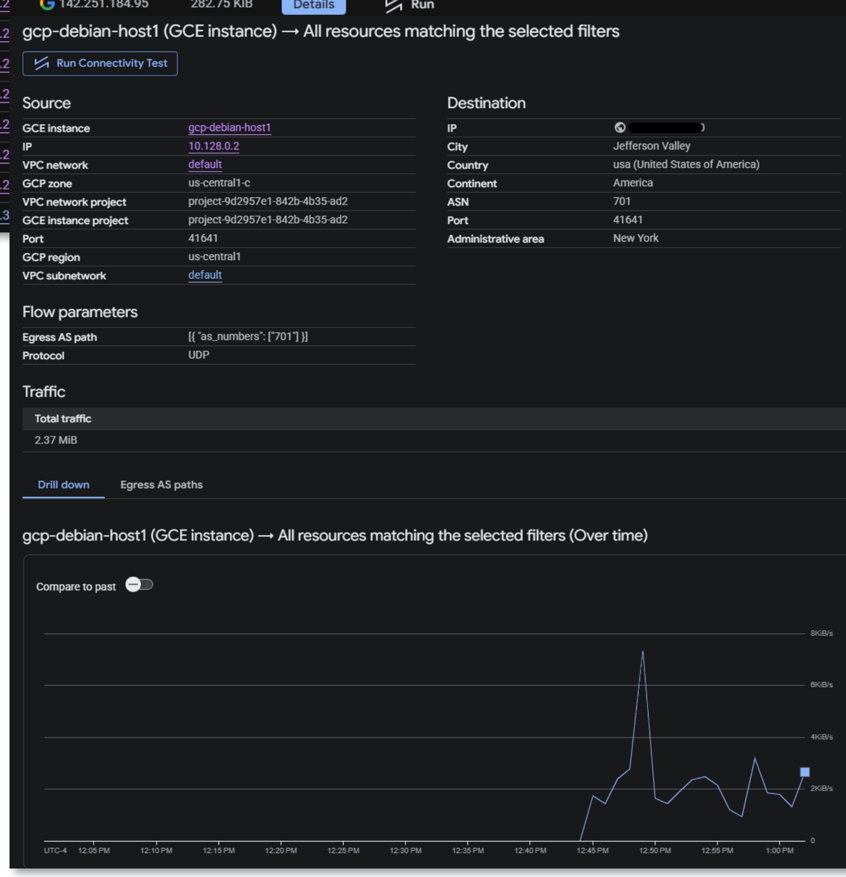
Cloud Monitoring

Infrastructure metrics (CPU, memory, disk, network) collected from GCP instances via the Cloud Monitoring agent (Ops Agent). Custom dashboards and alerting policies trigger notifications on resource thresholds. Complements on-premises Checkmk and Prometheus for cross-platform visibility.

VPC Flow Logs

The default network has flow logs enabled, connection history can be queried in Flow Analyzer.





IAM

GCP IAM controls access to all project resources. Service accounts with minimal roles are used for instance operations. No owner or editor roles assigned to compute instances. IAM policy recommendations surface over-permissive bindings. Workload Identity is planned for service-to-service authentication without key files.

Monitoring and Alerting

Source	Alert Type	Destination
Security Command Center	Misconfiguration findings, vulnerability detections	GCP console + log-based alerts
Cloud Armor	WAF blocks, rate limit triggers	Cloud Logging + alerting policies
Cloud Logging	High-risk API activity (IAM changes, firewall modifications)	Cloud Monitoring alerting policies
Cloud Monitoring	Instance resource thresholds	Cloud Monitoring alert + notification channel
Wazuh Agent	Endpoint security events	On-premises Wazuh Manager + SIEM
PatchMon Agent	Outdated packages / CVE correlation	On-premises PatchMon dashboard + Discord
Checkmk Agent	Host health (CPU, disk, services)	On-premises Checkmk + Discord
Uptime Kuma	Service availability checks	Discord webhook (#uptime-kuma)

Microsoft Azure

Deployment Overview

The Azure deployment consists of two VMs in the North Central US region: a Standard_ B2ats v2 instance running Ubuntu 24.04 LTS and a Standard_ E2s v3 Spot instance running Windows Server 2025 Datacenter Smalldisk. Both are enrolled in the Tailscale tailnet and provisioned via the azuredeploy.json ARM template. The Windows Server 2025 Core instance is joined to the home.com Active Directory domain over Tailscale, extending on-premises Group Policy, certificate auto-enrollment, and WSUS patch management to the cloud workload. Azure native services; Defender for Cloud, Log Analytics (homelab-log), Microsoft Sentinel, Entra ID, MFA, Azure Arc, and DDoS Protection (homelab-ddos) provide Microsoft-native security visibility that complements the on-premises SOC platform. The Azure VNet homelab-vnet2 (10.130.0.0/16) is in North Central US and is advertised into the Tailscale tailnet via the Ubuntu node acting as subnet router.

Security Impact

- Wazuh agents on both VMs provide endpoint telemetry forwarded to the central on-premises Wazuh Manager (192.168.1.219) via Tailscale
- Windows Server 2025 joined to home.com AD; Group Policy, WSUS, and certificate auto-enrollment operate across the Tailscale tunnel
- NSG enforces explicit inbound/outbound rules; management restricted to Tailscale CGNAT (100.64.0.0/10) and homelab subnets; all internet ingress denied at highest priority
- Defender for Cloud delivers CSPM posture management and workload protection with regulatory compliance benchmarking (CIS, NIST, PCI-DSS)
- Azure Arc integration applies consistent governance policies across Azure, AWS, and on-premises Proxmox nodes
- Microsoft Sentinel aggregates Azure Monitor logs with custom KQL detection rules and automated response playbooks
- Entra ID provides cloud identity with conditional access policies and MFA enforcement
- Log Analytics Workspace retains all audit and diagnostic logs with query-driven alerting

- On-premises Unbound DNS resolvers (192.168.1.153/154) provide consistent name resolution and malware domain filtering via Tailscale for the Ubuntu node; AD DNS used by the Windows domain member

Deployment Rationale

Azure completes the three-provider hybrid architecture and demonstrates proficiency with the Microsoft cloud security stack. Entra ID and Defender for Cloud are deployed in the majority of enterprise environments alongside AWS or GCP. The dual-VM topology adds a Windows Server 2025 domain member, mirroring hybrid identity patterns where cloud workloads participate in on-premises AD for centralized identity, Group Policy, and patch management. The Spot instance model demonstrates cost-optimized lab operations while maintaining full security control parity with on-demand deployments.

Architecture Principles Alignment

Defense in Depth

- Wazuh EDR coverage provides endpoint-layer visibility independent of Azure-native controls
- Defender for Cloud and Sentinel operate as independent detection layers above network controls
- Azure Arc extends on-premises Defender governance policies to cloud VMs

Secure by Design

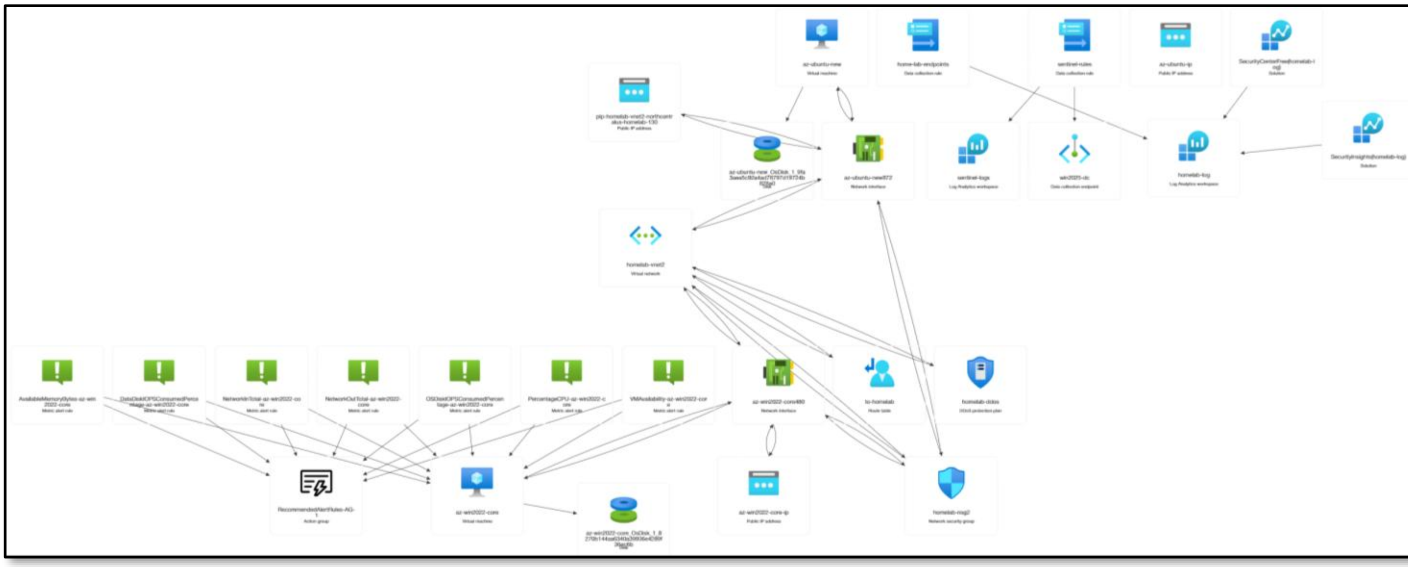
- VMs provisioned via secure ARM template parameters; adminPassword passed as securestring and never stored in template state
- NSG restricts SSH and RDP to 192.168.0.0/16 (homelab only); no public-internet management exposure
- Outbound rules explicitly permit only required management ports (Wazuh, DNS, Checkmk, PatchMon)
- Windows Server 2025 credential management via domain accounts, Kerberos, and WSUS-managed patches

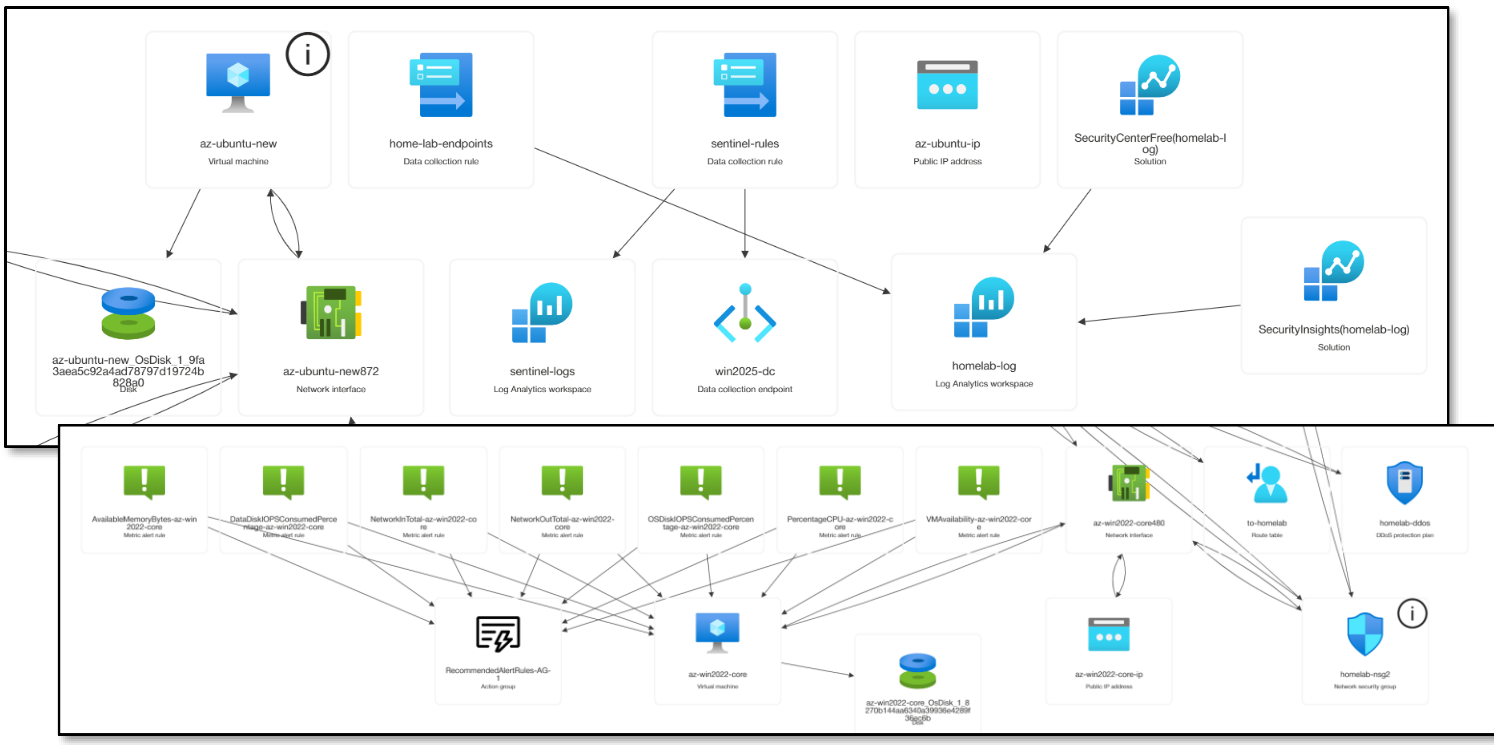
Zero Trust

- All management access via Tailscale, public IP assigned only for Tailscale UDP handshake
- NSG deny-by-default with explicit allow list mirrors pfSense and GCP firewall policy structure
- Entra ID conditional access enforces MFA for all administrative actions

Infrastructure Configuration

Resource Group: homelab





VM Instances

Ubuntu 24.04 LTS — az-ubuntu

Attribute	Value
VM Name	az-ubuntu
Resource Type	Microsoft.Compute/virtualMachines
VM Size	Standard B2ats v2
OS Offer	ubuntu-24_04-lts (Canonical)
SKU	server
Image Version	latest
Priority	On-Demand

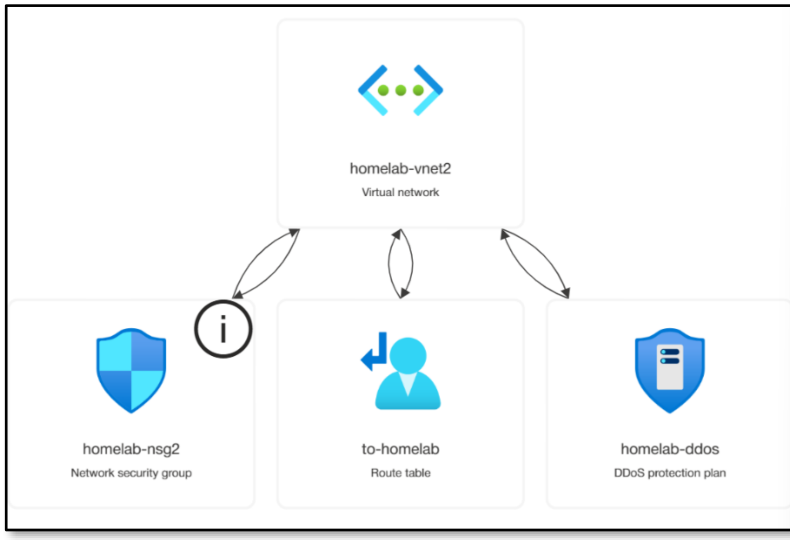
OS Disk	az-ubuntu_OsDisk_1_dd90488449a5407f97defc7ab67e2beb
NIC	az-ubuntu440
Admin Username	paul
Location	North Central US
Private IP	10.130.0.7
Public IP	az-ubuntu-ip
Tailscale IP	100.96.110.40
Tailscale Hostname	az-ubuntu.tailf07c05.ts.net
Tailscale Role	Subnet Router (10.130.0.0/16)/Exit Node
DNS Resolvers	192.168.1.153, 192.168.1.154 (on-premises Unbound via Tailscale)

Windows Server 2025 Datacenter Azure Edition— az-win2025-dc (note: original name az-win2022-core)

Attribute	Value
VM Name	az-win2025-dc
Resource Type	Microsoft.Compute/virtualMachines
VM Size	Standard E2s v3
OS Offer	WindowsServer 2025-datacenter-azure-edition-smalldisk (Microsoft)
Image Version	latest
Priority	Spot
Eviction Policy	Deallocate
OS Disk	az-win2025-dc_OsDisk_1_8270b144aa6340a39936e4289f36ec6b
NIC	az-win2025-dc480
Admin Username	Paul
Location	North Central US
Private IP	10.130.0.5
Public IP	az-win2025-dc-ip
Tailscale IP	100.113.13.112
Tailscale Hostname	az-win2025-dc.tailf07c05.ts.net
Tailscale Role	Endpoint
Domain	home.com (domain-joined via Tailscale)
DNS Resolvers	192.168.1.152, 192.168.1.142 (on-premises AD DNS via Tailscale — required for domain join)
Functions	domain member, Wazuh agent, Checkmk agent, WSUS client

VNet and Subnet Configuration

Attribute	Value
VNet Name	homelab-vnet2
VNet Address Space	10.130.0.0/16
Subnet Name	homelab-subnet
Subnet CIDR	10.130.0.0/24
NSG Attached	homelab-nsg2 (applied at subnet level, North Central US)
DDoS Protection	homelab-ddos (DDoS protection plan, North Central US)
Public IP (Ubuntu)	az-ubuntu-ip
Public IP (Windows)	az-win2025-dc-ip
NIC (Ubuntu)	az-ubuntu440
NIC (Windows)	az-win2025-dc480
Route Table	to-homelab



Azure Resource Inventory

All resources provisioned in the homelab resource group under Azure subscription 1. The majority are in North Central US; homelab-nsg and the East US 2 Network Watcher are legacy from the initial deployment.

Resource Name	Type	Location
az-ubuntu	Virtual machine	North Central US
az-ubuntu-ip	Public IP address	North Central US
az-ubuntu440	Network Interface	North Central US
az-ubuntu_OsDisk_1_dd90488449a5407f97defc7ab67e2beb	Disk	North Central US
az-win2025-dc	Virtual machine	North Central US
az-win2025-dc-ip	Public IP address	North Central US
az-win2025-dc480	Network Interface	North Central US
az-win2025-dc OsDisk_1_8270b144aa6340a39936e4289f36ec6b	Disk	North Central US
home-lab-endpoints	Data collection rule	North Central US
homelab-ddos	DDoS protection plan	North Central US
homelab-log	Log Analytics workspace	North Central US

homelab-nsg	Network security group	East US 2
homelab-nsg2	Network security group	North Central US
homelab-vnet2	Virtual network	North Central US
win2025-dc	Data collection endpoint	North Central US
to-homelab	Route table	North Central US
RecommendedAlertRules-AG-1	Action group	Global
SecurityCenterFree(homelab-log)	Solution	North Central US
NetworkWatcher_northcentralus	Network Watcher	North Central US
NetworkWatcher_eastus2	Network Watcher	East US 2
Available Memory Bytes - az-win2025-dc	Metric alert rule	Global
Data Disk IOPS Consumed Percentage - az-win2025-dc	Metric alert rule	Global
Network In Total - az-win2025-dc	Metric alert rule	Global
Network Out Total - az-win2025-dc	Metric alert rule	Global
OS Disk IOPS Consumed Percentage - az-win2025-dc	Metric alert rule	Global
Percentage CPU - az-win2025-dc	Metric alert rule	Global
VM Availability - az-win2025-dc	Metric alert rule	Global

Network Security Group — homelab-nsg2

homelab-nsg2 is the active NSG attached to the homelab-vnet2 subnet in North Central US. It enforces all inbound and outbound rules for both Azure VMs. The original homelab-nsg in East US 2 is retained from the initial deployment. Highest-priority rules deny threat sources before lower-priority allow rules are evaluated, mirroring the GCP VPC firewall policy and AWS security group structure.

Inbound Rules

Priority	Protocol	Port	Source	Action	Description
1000	*	*	100.64.0.0/10	Allow	Tailscale CGNAT — bidirectional Tailnet communication
1040	TCP	22	192.168.0.0/16	Allow	SSH — homelab subnets only (Ubuntu)
1044	*	6516	Any	Allow	Windows Admin Center Access
1045	TCP	3389	192.168.0.0/16	Allow	RDP — homelab subnets only (Windows)
1050	ICMP	*	192.168.0.0/16	Allow	ICMP diagnostics from homelab
1060	TCP	80	192.168.1.181/32	Allow	Uptime Kuma HTTP health checks
1070	UDP	33434-33600	192.168.0.0/16	Allow	Traceroute from homelab
65000	*	*	Any	Allow	VirtualNetwork: AllowVnetInBound
65001	*	*	Any	Allow	AzureLoadBalancer: AllowAzureLoadBalance
65500	*	*	Any	Deny	DenyAllInBound

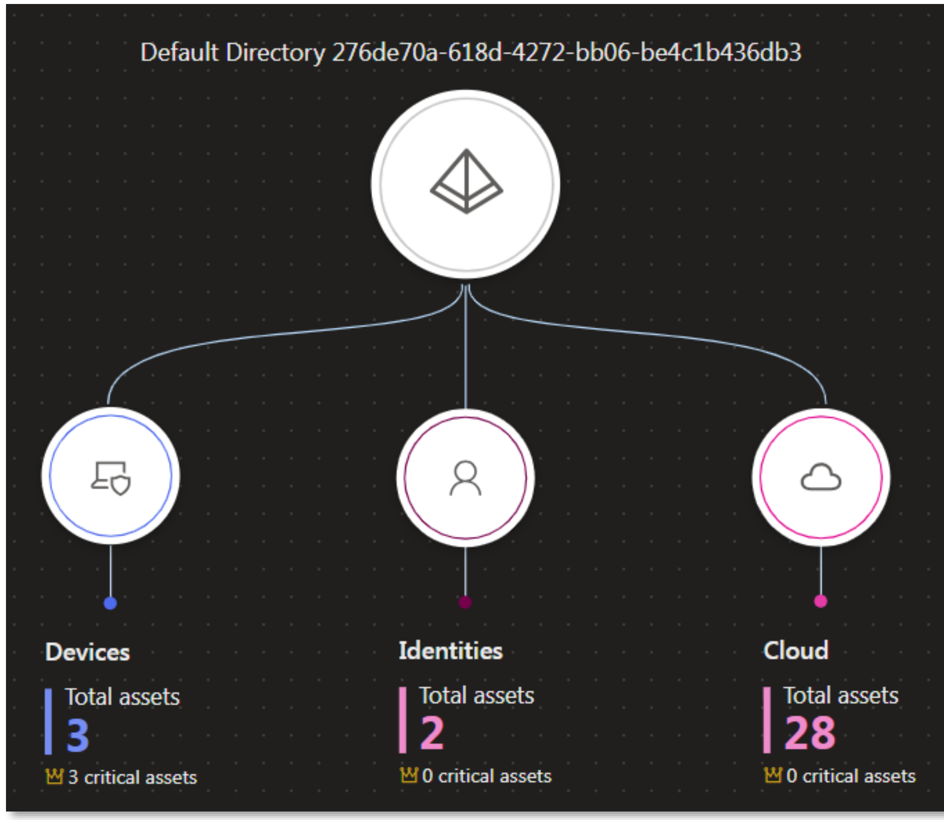
Outbound Rules

Priority	Protocol	Port(s)	Destination	Action	Description
100	*	*	Any	Allow	AzureCloud
2000	*	*	100.64.0.0/10	Allow	Tailscale outbound: all Tailnet destinations
2010	TCP	1514-55000	192.168.1.219/32	Allow	Wazuh agent manager event and enrollment ports
2020	*	53	192.168.1.153/32, 192.168.1.154/32	Allow	DNS: Unbound-01 and Unbound-02
2025	*	53	192.168.1.152/32, 192.168.1.142/32	Allow	DNS: AD DNS DC01 and DC02 (Windows domain join)
2030	TCP	*	192.168.1.126/32	Allow	Checkmk agent egress
2040	TCP	3000-3001	192.168.200.0/24	Allow	PatchMon backend
2050	All	AD Ports	192.168.1.152/32, 192.168.1.142/32	Allow	AD Domain Controller: Kerberos, LDAP, DNS, GP, SYSVOL
3000	TCP	443	WindowsAdminCenter	Allow	Windows Admin Center Access
3001	TCP	443	AzureActiveDirectory	Allow	Azure Active Directory Access
65000	*	*	Any	Allow	VirtualNetwork: AllowVnetOutBound
65001	*	*	Any	Allow	Internet: AllowInternetOutBound
65500	*	*	Any	Deny	DenyAlloutBound

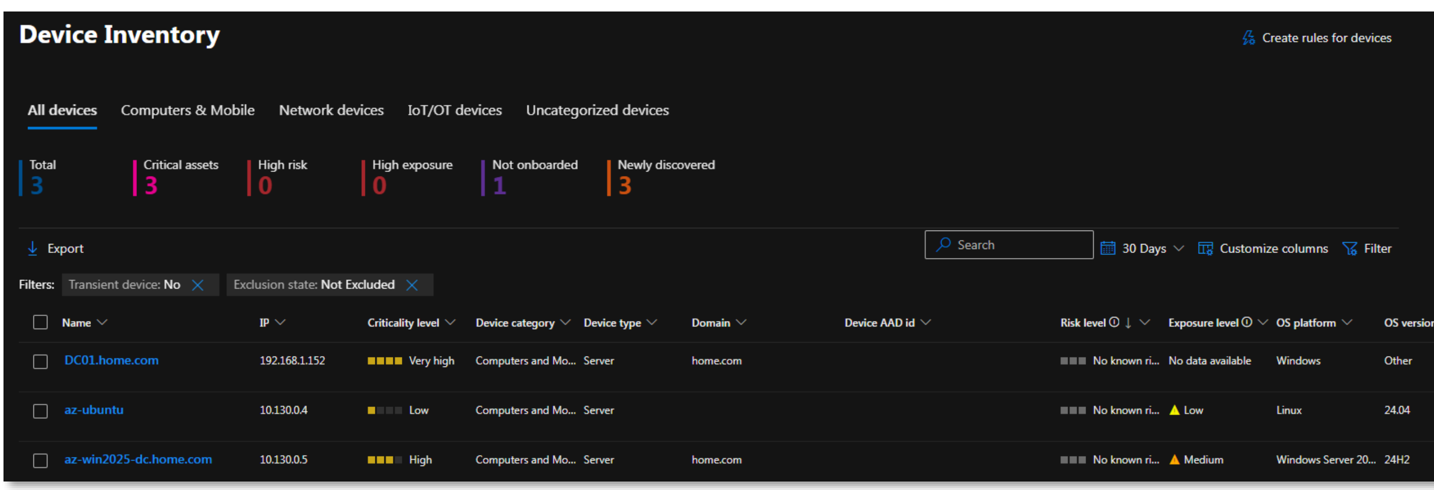
Native Security Services

Defender for Cloud

Defender for Cloud provides Cloud Security Posture Management (CSPM) across the Azure subscription. The lab configuration targets high visibility at zero or low cost through the following deployment choices.



- **Foundational CSPM (Free):** Enabled in Defender for Cloud > Environment Settings. Provides the Secure Score dashboard and compliance tracking against CIS Azure Foundations, PCI-DSS, and NIST SP 800-53 without per-node cost. Surfaces misconfigured NSG rules, missing MFA, public IP exposure, and unencrypted disks.
- **Azure Arc Integration:** The Connected Machine Agent (Arc) is installed on the Azure VMs and is planned for extension to Proxmox-hosted Linux VMs and AWS EC2 nodes. Arc enables Defender to apply the same governance policies and Secure Score recommendations across heterogeneous infrastructure — cloud-native, on-premises hypervisor, and multi-cloud — from a single control plane.
- **Defender for Servers Plan 1:** Enables EDR features and Microsoft Defender for Endpoint integration on enrolled VMs. Plan 1 provides vulnerability assessment and adaptive application controls. For this lab, the Wazuh EDR agent serves as the primary endpoint detection layer; Defender for Servers is deployed as a supplementary control when trial capacity is available.

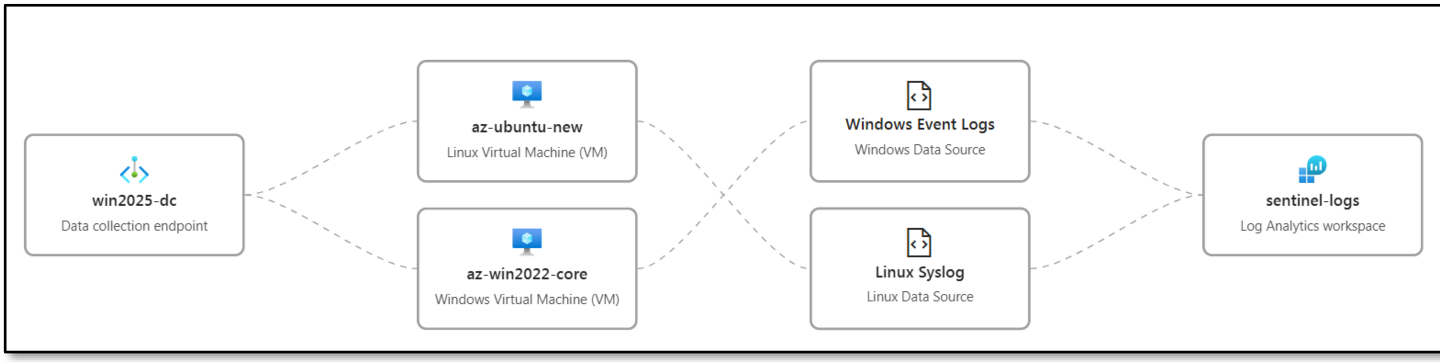


Microsoft Sentinel

Microsoft Sentinel serves as the cloud-native SIEM/SOAR for the Azure deployment and aggregates data from across the hybrid environment. The Log Analytics Workspace (homelab-log) is the central repository; Sentinel is linked to it at creation time.

Configuration steps applied in the lab:

1. Create Log Analytics Workspace: homelab-log is provisioned in North Central US. The workspace uses the Pay-As-You-Go commitment tier with a 90-day retention window, matching the on-premises SIEM retention policy.
2. Enable Sentinel: Microsoft Sentinel is enabled from the Azure Portal by linking to homelab-log. Analytics rules, data connectors, and automation rules are configured post-enablement.



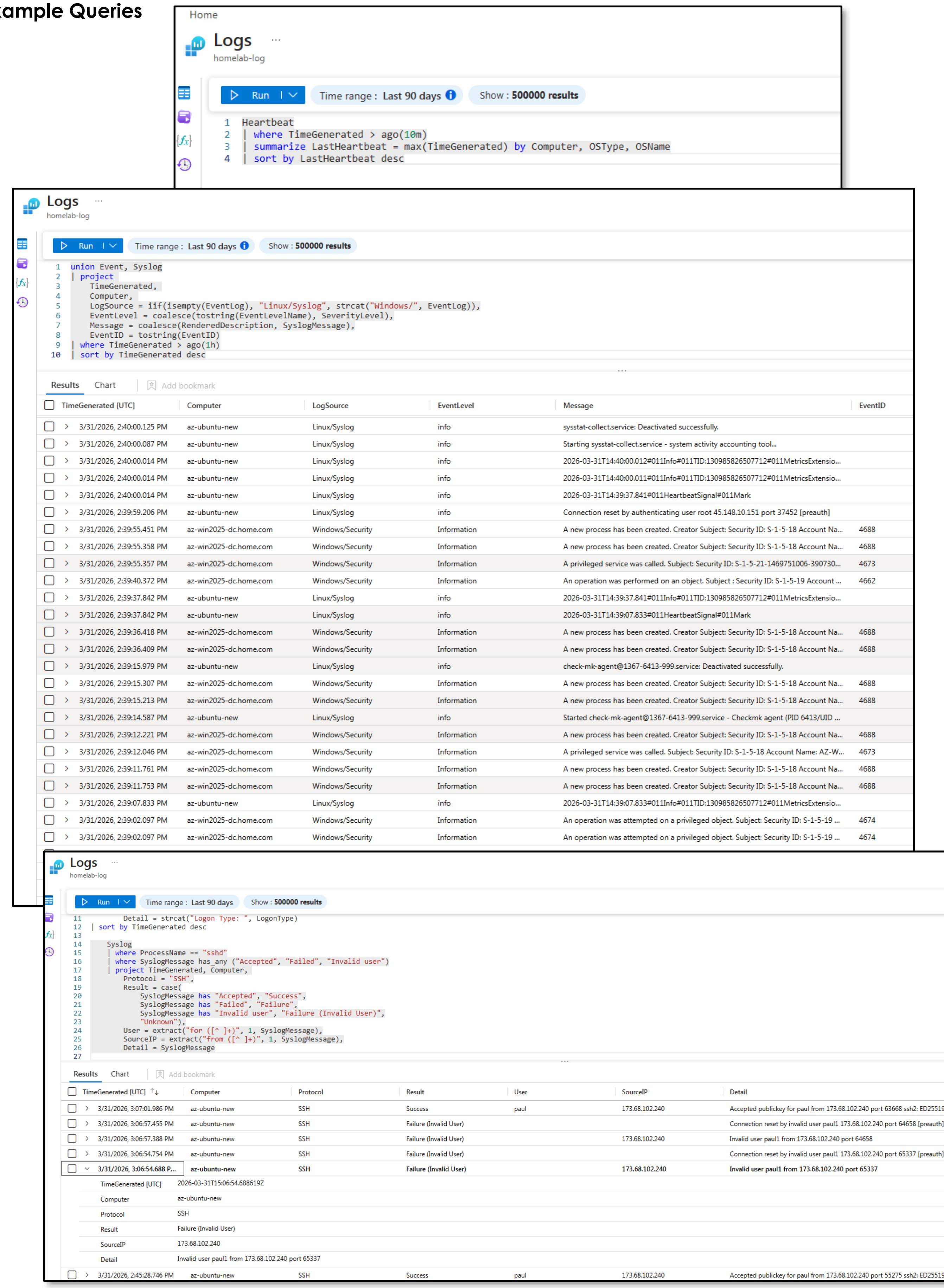
Data Connectors (low-cost focus):

- **Azure Activity (Free):** Streams all ARM template deployments, resource changes, and control-plane operations from the Azure subscription into Sentinel. Provides visibility into NSG rule modifications, VM provisioning events, and IAM role assignments without additional cost.
- **Microsoft Entra ID (Free tier):** Ingests Audit Logs and Sign-in Logs from Entra ID. Essential for tracking home.com synced user activity, conditional access policy hits, MFA challenge results, and privileged account sign-ins across the Azure tenant.
- **Windows Security Events via AMA:** The Azure Monitor Agent (AMA) is deployed via data collection rule home-lab-endpoints (data collection endpoint: win2025-dc). The DCR targets the Common event set (4624, 4625, 4648, 4768, 4769, 4771, 7045, 7040, 4720, 4728, 4732, 4756) and streams events into homelab-log from all enrolled resources. This provides cross-cloud Windows event correlation within Sentinel without requiring full Security Events (which incurs higher ingestion cost).
- **Syslog via AMA (Linux nodes):** Syslog forwarding is configured on az-ubuntu and gcp-debian-host1 via the home-lab-endpoints DCR. The AMA collects auth, syslog, and daemon facility events. Combined with the Wazuh agent forwarding to the on-premises SIEM, this provides dual-path visibility for Linux security events.

KQL Detection Rules (custom, deployed in the lab):

- **NSG Deny-Inbound Spike:** Alerts when the homelab-nsg2 priority-100 deny rule fires more than 20 times per hour from a single source IP, indicating active scanning or a blocked attack.
- **Entra ID Sign-in from New Location:** Correlates Entra ID sign-in logs against a known-good location baseline. Fires when an administrative account authenticates from a country not seen in the prior 30 days.
- **Windows Domain Join Event:** Monitors Security Event 4742 (computer account changed) and correlates with the expected domain-join window. Unexpected domain joins outside provisioning windows generate a Sentinel incident.
- **AMA DCR Data Gap:** Fires when the AMA heartbeat for any monitored VM is absent for more than 15 minutes, indicating agent failure or VM eviction.

Example Queries

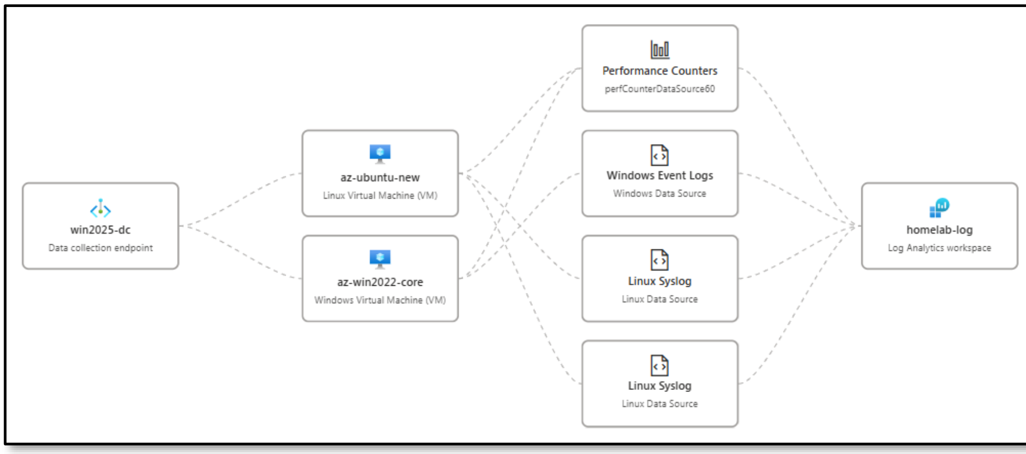


Entra ID and MFA

Entra ID serves as the cloud identity provider for Azure resource access. Conditional access policies enforce MFA for all administrative roles. Privileged Identity Management (PIM) provides just-in-time role activation for subscription-level permissions. Sign-in logs and audit events are forwarded to homelab-log for retention and correlation with Sentinel analytics rules.

Log Analytics Workspace

homelab-log is the central log repository for all Azure diagnostic data: VM performance counters, NSG flow logs, Defender for Cloud alerts, Sentinel incidents, and Entra ID sign-in events. The



SecurityCenterFree(homelab-log) solution extends Defender for Cloud data into the workspace. KQL queries provide ad-hoc investigation capability. Alert rules (action group: RecommendedAlertRules-AG-1) forward critical events to notification channels and Sentinel automation rules that can trigger Shuffle SOAR webhooks on the on-premises platform.

Azure Arc

The Connected Machine Agent is installed on both Azure VMs and is planned for extension to on-premises Proxmox-hosted Linux hosts and AWS EC2 instances. Arc registers each machine as a resource in Azure Resource Manager, enabling Defender for Cloud Secure Score recommendations, Azure Policy assignments, and Guest Configuration auditing to apply uniformly across cloud and on-premises workloads. This provides a single compliance dashboard spanning the full hybrid environment.

Monitoring and Alerting

Source	Alert Type	Destination
Defender for Cloud	Misconfiguration findings, vulnerability detections	Azure portal + Log Analytics alerts
Microsoft Sentinel	Detection rule hits, incident creation	Automation rules + Shuffle SOAR webhook
NSG Flow Logs	Anomalous traffic patterns	Log Analytics alerting + Sentinel
Log Analytics	VM diagnostic thresholds	Azure Monitor alerts
Wazuh Agent	Endpoint security events	On-premises Wazuh Manager + Splunk SIEM
PatchMon Agent	Outdated packages / CVE correlation	On-premises PatchMon dashboard + Discord
Checkmk Agent	Host health (CPU, disk, services)	On-premises Checkmk + Discord

Uptime Kuma	Service availability checks	Discord webhook (#uptime-kuma)

On-Premises Management Integration

All cloud nodes — across AWS, GCP, and Azure — are managed using the same on-premises toolchain applied to Proxmox-hosted workloads. This provides a single operational model regardless of infrastructure location: unified patch compliance dashboards, consistent security baselines, centralized alerting, and a common incident response pipeline. The toolchain reaches cloud nodes exclusively via the Tailscale mesh VPN. No management ports are exposed to the public internet.

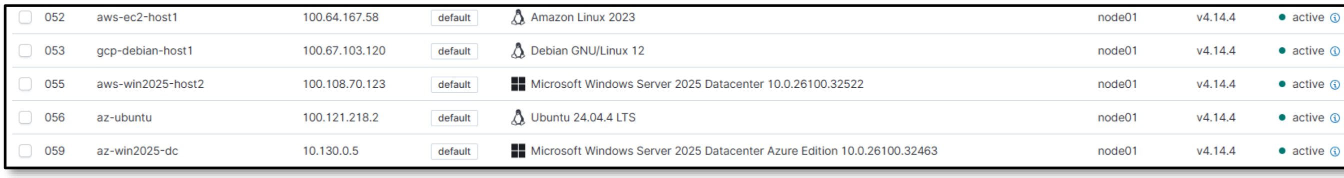
Tool	Function	Reach	Port / Protocol
Wazuh EDR	Endpoint detection, FIM, SCA, vulnerability assessment	All cloud nodes	TCP 1514/1515 outbound to 192.168.1.219
PatchMon	Package currency tracking, CVE correlation, patch SLA tracking	Linux cloud nodes	TCP 3000-3001 outbound to 192.168.200.0/24
Checkmk	Host health monitoring (CPU, memory, disk, services)	All cloud nodes	TCP 6556 inbound from 192.168.1.126
Ansible	Configuration baselines, hardening, user management, patching	All cloud nodes	SSH (TCP 22) inbound from 192.168.1.25
Uptime Kuma	Service availability and tunnel health checks	All cloud nodes	TCP 22/3389 inbound from 192.168.1.181
WSUS	Windows patch management	Windows cloud nodes	HTTP/HTTPS outbound to 192.168.1.152

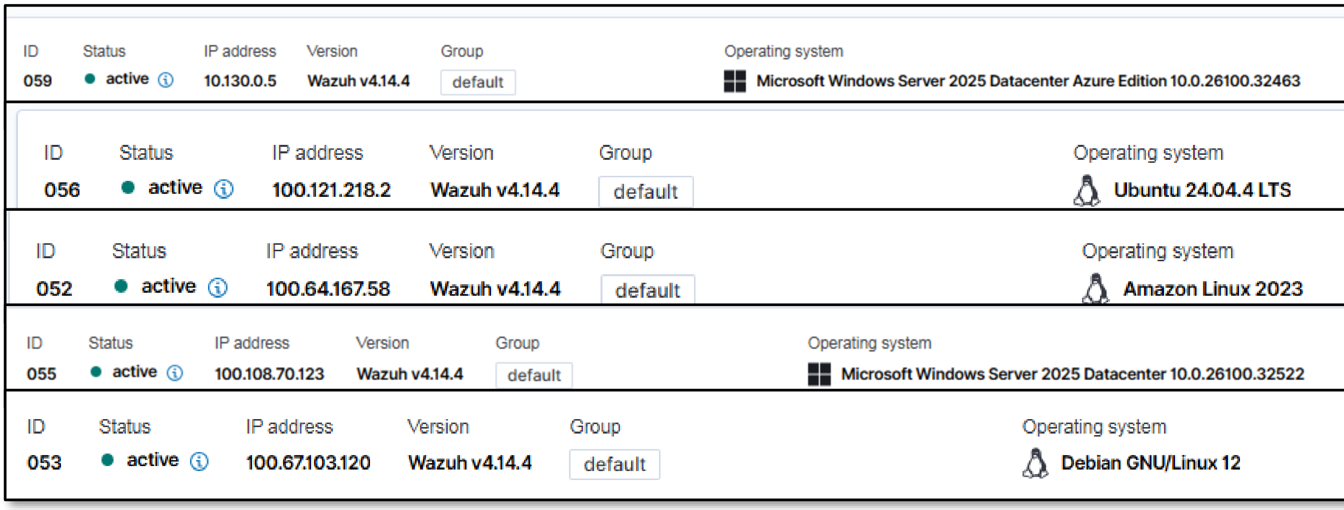
Wazuh EDR

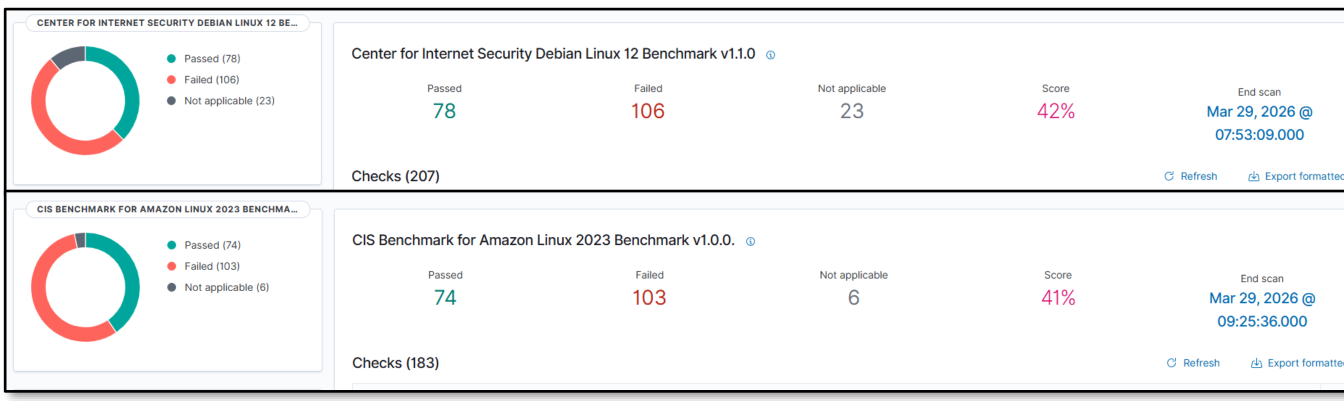
Wazuh agents are deployed on all cloud instances and report to the on-premises Wazuh Manager (192.168.1.219) via the Tailscale tunnel. The agent provides file integrity monitoring, rootkit detection, vulnerability assessment, CIS benchmark Security Configuration Assessment (SCA), and real-time security event forwarding to the on-premises SIEM (Splunk + Elastic).

Parameter	Value
Wazuh Manager	192.168.1.219 (on-premises)
Enrollment Port	TCP 1515
Event Forwarding Port	TCP 1514
API Port	TCP 55000
Transport	Tailscale tunnel (WireGuard encrypted)
Traffic Flow	Cloud Agent → Tailscale → pfSense → Wazuh Manager (192.168.1.219)
SCA Policies Applied	CIS Amazon Linux, CIS Debian 12/13, CIS Ubuntu 22.04/24.04, CIS Windows Server 2025, CIS RHEL 10
Active Response	firewall-drop, host-deny, disable-account on threshold breach

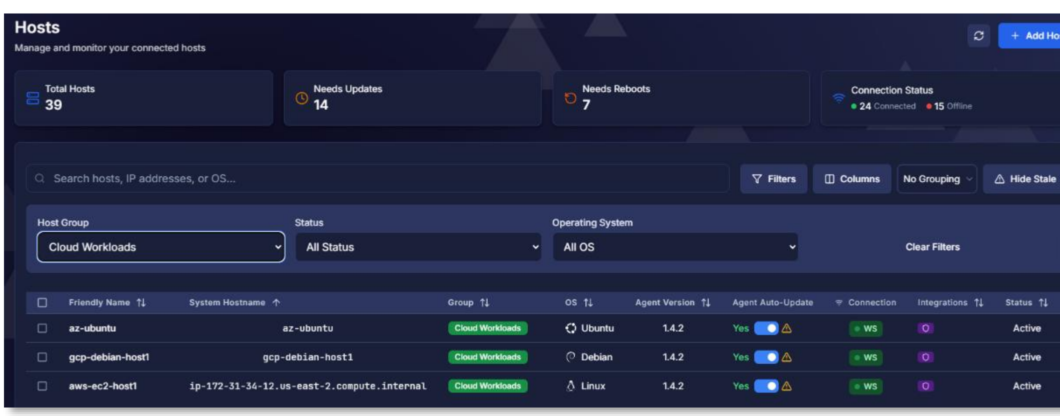
Cloud nodes appear as named agents in the Wazuh dashboard alongside on-premises hosts. SCA compliance scores, vulnerability counts, and FIM alerts are visible in the unified Wazuh and Splunk dashboards without platform-specific configuration.





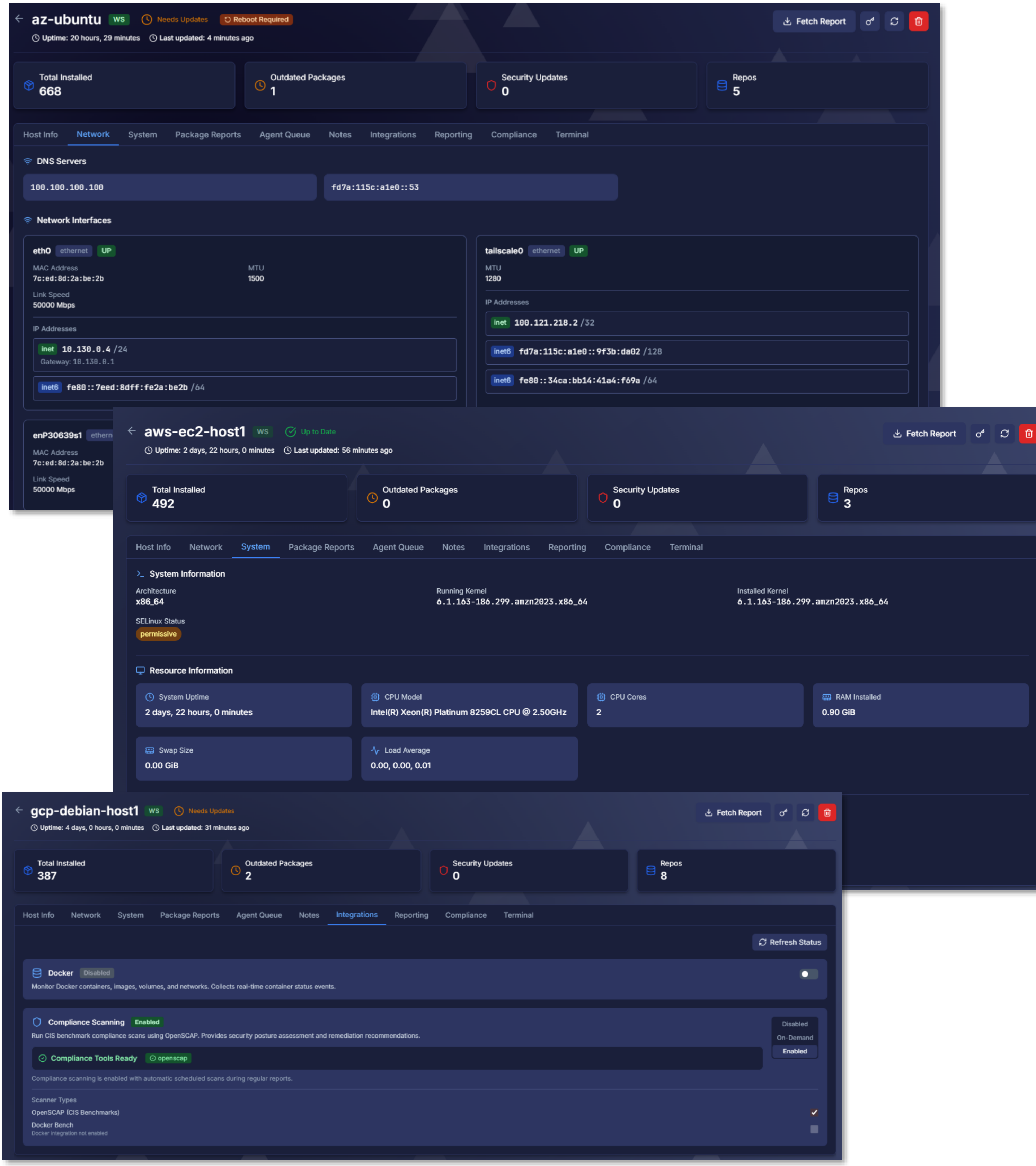


PatchMon



PatchMon agents are deployed on all Linux cloud nodes (Amazon Linux 2, Debian 13.4, Ubuntu 24.04 LTS). The agent polls the native package manager (yum/dnf for Amazon Linux and RHEL-family; apt for Debian/Ubuntu) on a daily schedule and reports available updates to the PatchMon backend (192.168.200.39) via Tailscale.

Package versions are correlated against the NVD database to identify CVE-linked updates and calculate per-host security update counts.



Windows cloud nodes (AWS Windows Server 2025, Azure Windows Server 2025 DC) are managed via WSUS (192.168.1.152) for Microsoft product updates. PatchMon does not monitor Windows endpoints; WSUS compliance data is visible in the PatchMon dashboard via the Windows Update Services integration.

All cloud hosts appear in the unified PatchMon dashboard alongside on-premises hosts, grouped by cloud provider. Critical CVE MTTR target is maintained at <72 hours across all platforms.

Checkmk

Checkmk agents are deployed on all cloud instances and report to the on-premises Checkmk server (192.168.1.126) via Tailscale. The agent provides OS-level metrics — CPU utilization, memory, disk usage, running services, and process health — consistent with the monitoring applied to on-premises Proxmox guests.

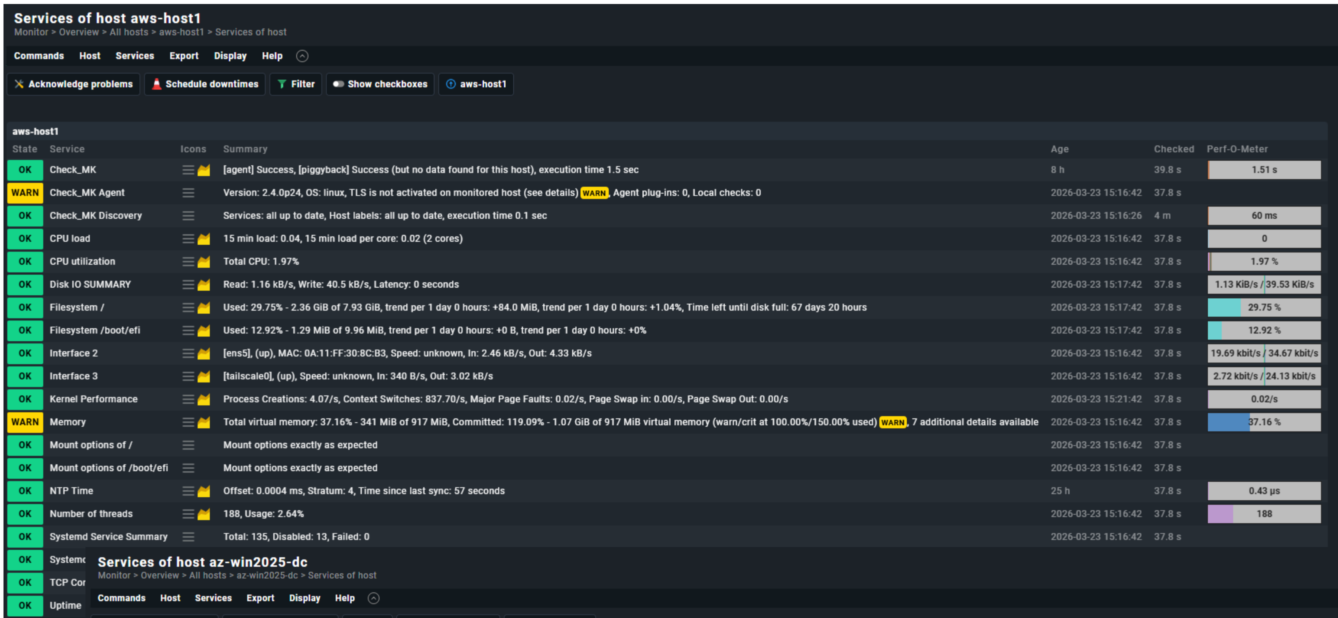
Node	Integration Method	Key Metrics
aws-ec2-host1 (Amazon Linux)	Checkmk Agent (TCP 6556)	CPU, memory, disk, systemd services, network interfaces
aws-win2025-host2 (Windows)	Checkmk Windows Agent (TCP 6556)	CPU, memory, disk, Windows services, event log summary
gcp-debian-host1 (Debian)	Checkmk Agent (TCP 6556)	CPU, memory, disk, systemd services, package count
gcp-ubuntu-host2 (Ubuntu)	Checkmk Agent (TCP 6556)	CPU, memory, disk, systemd services, package count
az-ubuntu (Ubuntu)	Checkmk Agent (TCP 6556)	CPU, memory, disk, systemd services
az-win2025-dc (Windows)	Checkmk Windows Agent (TCP 6556)	CPU, memory, disk, AD services (if applicable), Windows services

Hosts

Actions	Host name	Alias	IPv4 address
	Ansible		192.168.1.25
	aws-host1	AWS Amazon Linux	172.31.34.12
	aws-win2025-h2	AWS Windows Server 2025 Datacenter	172.31.46.238
	az-ubuntu	Azure Ubuntu 24.04 LTS	10.130.0.4
	az-win2025-dc	Azure Windows Server 2025 Datacenter Azure Edition	10.130.0.5
	gcp-debian-host1	Google Cloud Platform Debian 12	10.128.0.2

Local site cmk

State	Host	Icons	OK	Wa	Un	Cr
UP	Ansible		13	1	0	1
UP	aws-host1		18	2	0	0
UP	aws-win2025-h2		17	1	0	3
UP	az-ubuntu		22	2	0	0
UP	az-win2025-dc		21	1	0	1
DOWN	BentoPDF		11	2	0	2
UP	CentOS		11	1	0	1
DOWN	Cisco-r1		8	1	0	2
UP	Cisco-r2		9	1	0	0
DOWN	Cisco_Host		11	1	0	4
DOWN	Cisco_Host_2		11	1	0	4
UP	CrowdSec		12	2	0	1
UP	DC01-Server2022		14	2	0	9
UP	DC02-Win2025Server		18	2	0	7
UP	Debian-Docker2		17	2	2	3
DOWN	elastic		0	0	0	3
DOWN	ESXi-host1		0	0	0	3
UP	Fedora-Server		15	4	0	0
UP	gcp-debian-host1		16	2	0	2



Services of host aws-host1

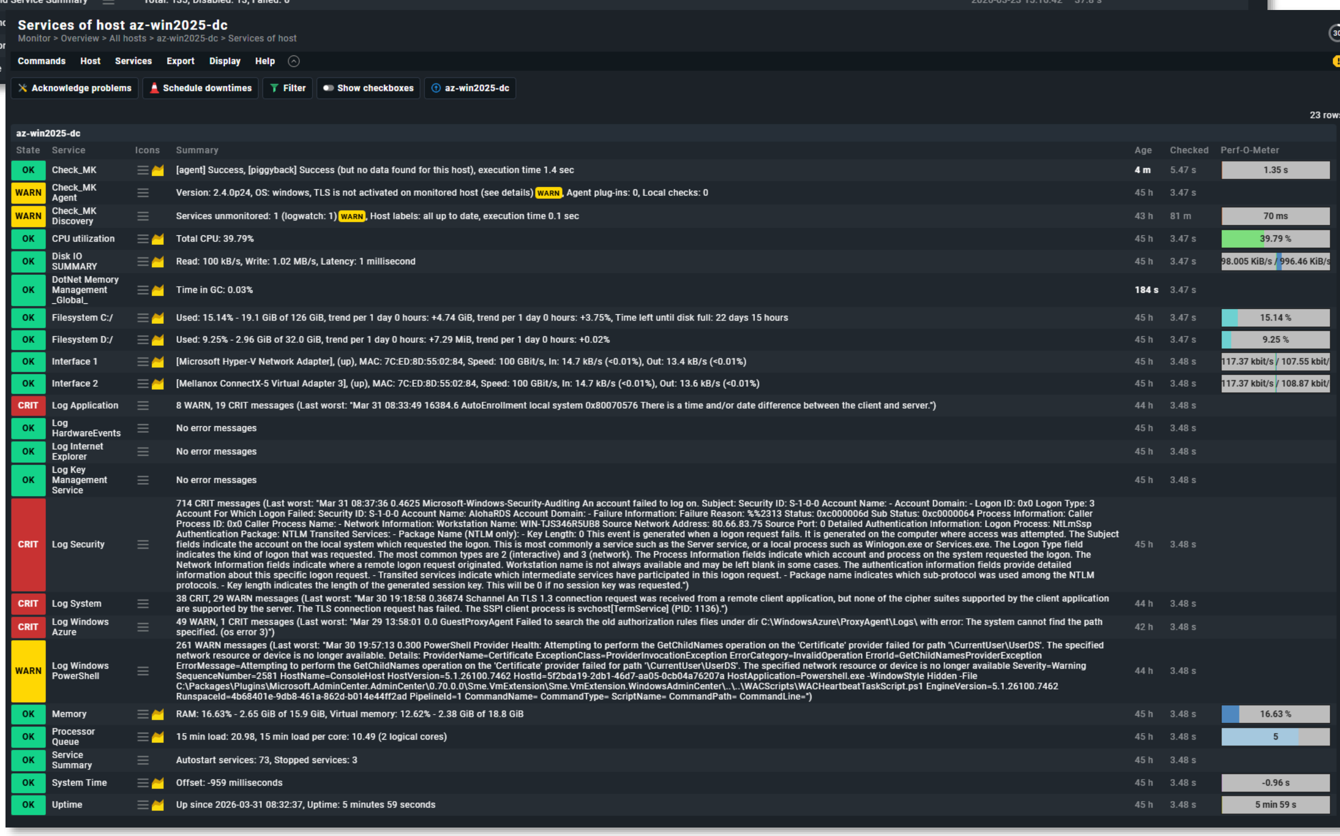
Monitor > Overview > All hosts > aws-host1 > Services of host

Commands Host Services Export Display Help

Acknowledge problems | Schedule downtimes | Filter | Show checkboxes | aws-host1

aws-host1

State	Service	Summary	Age	Checked	Perf-O-Meter
OK	Check_MK	[agent] Success, [piggyback] Success (but no data found for this host), execution time 1.5 sec	8 h	39.8 s	1.51 s
WARN	Check_MK Agent	Version: 2.4.0p24, OS: linux, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	2026-03-23 15:16:42	37.8 s	
OK	Check_MK Discovery	Services: all up to date, Host labels: all up to date, execution time 0.1 sec	2026-03-23 15:16:26	4 m	60 ms
OK	CPU load	15 min load: 0.04, 15 min load per core: 0.02 (2 cores)	2026-03-23 15:16:42	37.8 s	0
OK	CPU utilization	Total CPU: 1.97%	2026-03-23 15:16:42	37.8 s	1.97 %
OK	Disk IO SUMMARY	Read: 1.16 kB/s, Write: 40.5 kB/s, Latency: 0 seconds	2026-03-23 15:17:42	37.8 s	1.13 KiB/s / 39.53 KiB/s
OK	Filesystem /	Used: 29.75% - 2.36 GiB of 7.93 GiB, trend per 1 day 0 hours: +84.0 MiB, trend per 1 day 0 hours: +1.04%, Time left until disk full: 67 days 20 hours	2026-03-23 15:17:42	37.8 s	29.75 %
OK	Filesystem /boot/efi	Used: 12.92% - 1.29 MiB of 9.96 MiB, trend per 1 day 0 hours: +0 B, trend per 1 day 0 hours: +0%	2026-03-23 15:17:42	37.8 s	12.92 %
OK	Interface 2	[ens5], (up), MAC: 0A:11:FF:30:8C:B3, Speed: unknown, In: 2.46 kB/s, Out: 4.33 kB/s	2026-03-23 15:16:42	37.8 s	19.69 kbit/s / 34.67 kbit/s
OK	Interface 3	[tailscale0], (up), Speed: unknown, In: 340 B/s, Out: 3.02 kB/s	2026-03-23 15:16:42	37.8 s	2.72 kbit/s / 24.13 kbit/s
OK	Kernel Performance	Process Creations: 4.07/s, Context Switches: 837.70/s, Major Page Faults: 0.02/s, Page Swap in: 0.00/s, Page Swap Out: 0.00/s	2026-03-23 15:21:42	37.8 s	0.02/s
WARN	Memory	Total virtual memory: 37.16% - 341 MiB of 917 MiB, Committed: 119.09% - 1.07 GiB of 917 MiB virtual memory (warn/crit at 100.00%/150.00% used) WARN, 7 additional details available	2026-03-23 15:16:42	37.8 s	37.16 %
OK	Mount options of /	Mount options exactly as expected	2026-03-23 15:16:42	37.8 s	
OK	Mount options of /boot/efi	Mount options exactly as expected	2026-03-23 15:16:42	37.8 s	
OK	NTP Time	Offset: 0.0004 ms, Stratum: 4, Time since last sync: 57 seconds	25 h	37.8 s	0.43 µs
OK	Number of threads	188, Usage: 2.64%	2026-03-23 15:16:42	37.8 s	188
OK	Systemd Service Summary	Total: 135, Disabled: 13, Failed: 0	2026-03-23 15:16:42	37.8 s	
OK	Systemc				
OK	TCP Cor				
OK	Uptime				



Services of host az-win2025-dc

Monitor > Overview > All hosts > az-win2025-dc > Services of host

Commands Host Services Export Display Help

Acknowledge problems | Schedule downtimes | Filter | Show checkboxes | az-win2025-dc

23 rows

az-win2025-dc

State	Service	Summary	Age	Checked	Perf-O-Meter
OK	Check_MK	[agent] Success, [piggyback] Success (but no data found for this host), execution time 1.4 sec	4 m	5.47 s	1.35 s
WARN	Check_MK Agent	Version: 2.4.0p24, OS: windows, TLS is not activated on monitored host (see details) WARN, Agent plug-ins: 0, Local checks: 0	45 h	3.47 s	
WARN	Check_MK Discovery	Services unmonitored: 1 (logwatch: 1) WARN, Host labels: all up to date, execution time 0.1 sec	43 h	81 m	70 ms
OK	CPU utilization	Total CPU: 39.79%	45 h	3.47 s	39.79 %
OK	Disk IO SUMMARY	Read: 100 kB/s, Write: 1.02 MB/s, Latency: 1 millisecond	45 h	3.47 s	98.005 KiB/s / 996.46 KiB/s
OK	DotNet Memory Management _Global_	Time in GC: 0.03%	184 s	3.47 s	
OK	Filesystem C:/	Used: 15.14% - 19.1 GiB of 126 GiB, trend per 1 day 0 hours: +4.74 GiB, trend per 1 day 0 hours: +3.75%, Time left until disk full: 22 days 15 hours	45 h	3.47 s	15.14 %
OK	Filesystem D:/	Used: 9.25% - 2.96 GiB of 32.0 GiB, trend per 1 day 0 hours: +7.29 MiB, trend per 1 day 0 hours: +0.02%	45 h	3.47 s	9.25 %
OK	Interface 1	[Microsoft Hyper-V Network Adapter], (up), MAC: 7C:ED:8D:55:02:84, Speed: 100 GBit/s, In: 14.7 kB/s (<0.01%), Out: 13.4 kB/s (<0.01%)	45 h	3.48 s	117.37 kbit/s / 107.55 kbit/
OK	Interface 2	[Mellanox ConnectX-5 Virtual Adapter 3], (up), MAC: 7C:ED:8D:55:02:84, Speed: 100 GBit/s, In: 14.7 kB/s (<0.01%), Out: 13.6 kB/s (<0.01%)	45 h	3.48 s	117.37 kbit/s / 108.87 kbit/
CRIT	Log Application	8 WARN, 19 CRIT messages (Last worst: "Mar 31 08:33:49 16384.6 AutoEnrollment local system 0x80070576 There is a time and/or date difference between the client and server.")	44 h	3.48 s	
OK	Log HardwareEvents	No error messages	45 h	3.48 s	
OK	Log Internet Explorer	No error messages	45 h	3.48 s	
OK	Log Key Management Service	No error messages	45 h	3.48 s	
CRIT	Log Security	714 CRIT messages (Last worst: "Mar 31 08:37:36 0.4625 Microsoft-Windows-Security-Auditing An account failed to log on. Subject: Security ID: S-1-0-0 Account Name: - Account Domain: - Logon ID: 0x0 Logon Type: 3 Account For Which Logon Failed: Security ID: S-1-0-0 Account Name: AlohaRDS Account Domain: - Failure Information: Failure Reason: %%2313 Status: 0xc000006d Sub Status: 0xc0000064 Process Information: Caller Process ID: 0x0 Caller Process Name: - Network Information: Workstation Name: WIN-TJS346R5UB8 Source Network Address: 80.66.83.75 Source Port: 0 Detailed Authentication Information: Logon Process: NtLmSsp Authentication Package: NTLM Transited Services: - Package Name (NTLM only): - Key Length: 0 This event is generated when a logon request fails. It is generated on the computer where access was attempted. The Subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe. The Logon Type field indicates the kind of logon that was requested. The most common types are 2 (interactive) and 3 (network). The Process Information fields indicate which account and process on the system requested the logon. The Network Information fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases. The authentication information fields provide detailed information about this specific logon request. - Transited services indicate which intermediate services have participated in this logon request. - Package name indicates which sub-protocol was used among the NTLM protocols. - Key length indicates the length of the generated session key. This will be 0 if no session key was requested.")	45 h	3.48 s	
CRIT	Log System	38 CRIT, 29 WARN messages (Last worst: "Mar 30 19:18:58 0.36874 Schannel An TLS 1.3 connection request was received from a remote client application, but none of the cipher suites supported by the client application are supported by the server. The TLS connection request has failed. The SSPI client process is svchost[TermService] (PID: 1136).")	44 h	3.48 s	
CRIT	Log Windows Azure	49 WARN, 1 CRIT messages (Last worst: "Mar 29 13:58:01 0.0 GuestProxyAgent Failed to search the old authorization rules files under dir C:\WindowsAzure\ProxyAgent\Logs\ with error: The system cannot find the path specified. (os error 3)")	42 h	3.48 s	
WARN	Log Windows PowerShell	261 WARN messages (Last worst: "Mar 30 19:57:13 0.300 PowerShell Provider Health: Attempting to perform the GetChildNames operation on the 'Certificate' provider failed for path '\CurrentUser\UserDS'. The specified network resource or device is no longer available. Details: ProviderName=Certificate ExceptionClass=ProviderInvocationException ErrorCategory=InvalidOperation ErrorId=GetChildNamesProviderException ErrorMessage=Attempting to perform the GetChildNames operation on the 'Certificate' provider failed for path '\CurrentUser\UserDS'. The specified network resource or device is no longer available Severity=Warning SequenceNumber=2581 HostName=ConsoleHost HostVersion=5.1.26100.7462 HostId=5f2bda19-2db1-46d7-aa05-0cb04a76207a HostApplication=Powershell.exe -WindowStyle Hidden -File C:\Packages\Plugins\Microsoft.AdminCenter.AdminCenter\0.70.0.0\Sme.VmExtension\Sme.VmExtension.WindowsAdminCenter\..\..\WACScripts\WACHeartbeatTaskScript.ps1 EngineVersion=5.1.26100.7462 RunspaceId=4b68401e-9db8-461a-862d-b014e44ff2ad PipelineId=1 CommandName= CommandType= ScriptName= CommandPath= CommandLine=")	44 h	3.48 s	
OK	Memory	RAM: 16.63% - 2.65 GiB of 15.9 GiB, Virtual memory: 12.62% - 2.38 GiB of 18.8 GiB	45 h	3.48 s	16.63 %
OK	Processor Queue	15 min load: 20.98, 15 min load per core: 10.49 (2 logical cores)	45 h	3.48 s	5
OK	Service Summary	Autostart services: 73, Stopped services: 3	45 h	3.48 s	
OK	System Time	Offset: -959 milliseconds	45 h	3.48 s	-0.96 s
OK	Uptime	Up since 2026-03-31 08:32:37, Uptime: 5 minutes 59 seconds	45 h	3.48 s	5 min 59 s

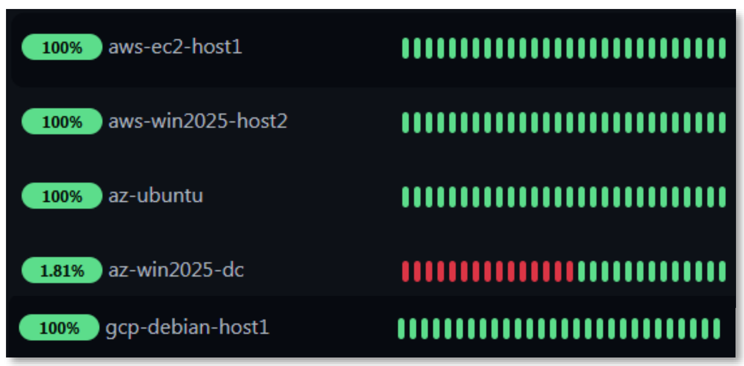
Ansible

All cloud Linux nodes are included in the Ansible inventory and managed from the Ansible controller (192.168.1.25) via SSH over Tailscale. Windows nodes are managed via WinRM. The same playbook library applied to on-premises hosts is reused across cloud workloads with OS-family detection handling Debian/Ubuntu versus RHEL/Amazon Linux differences.

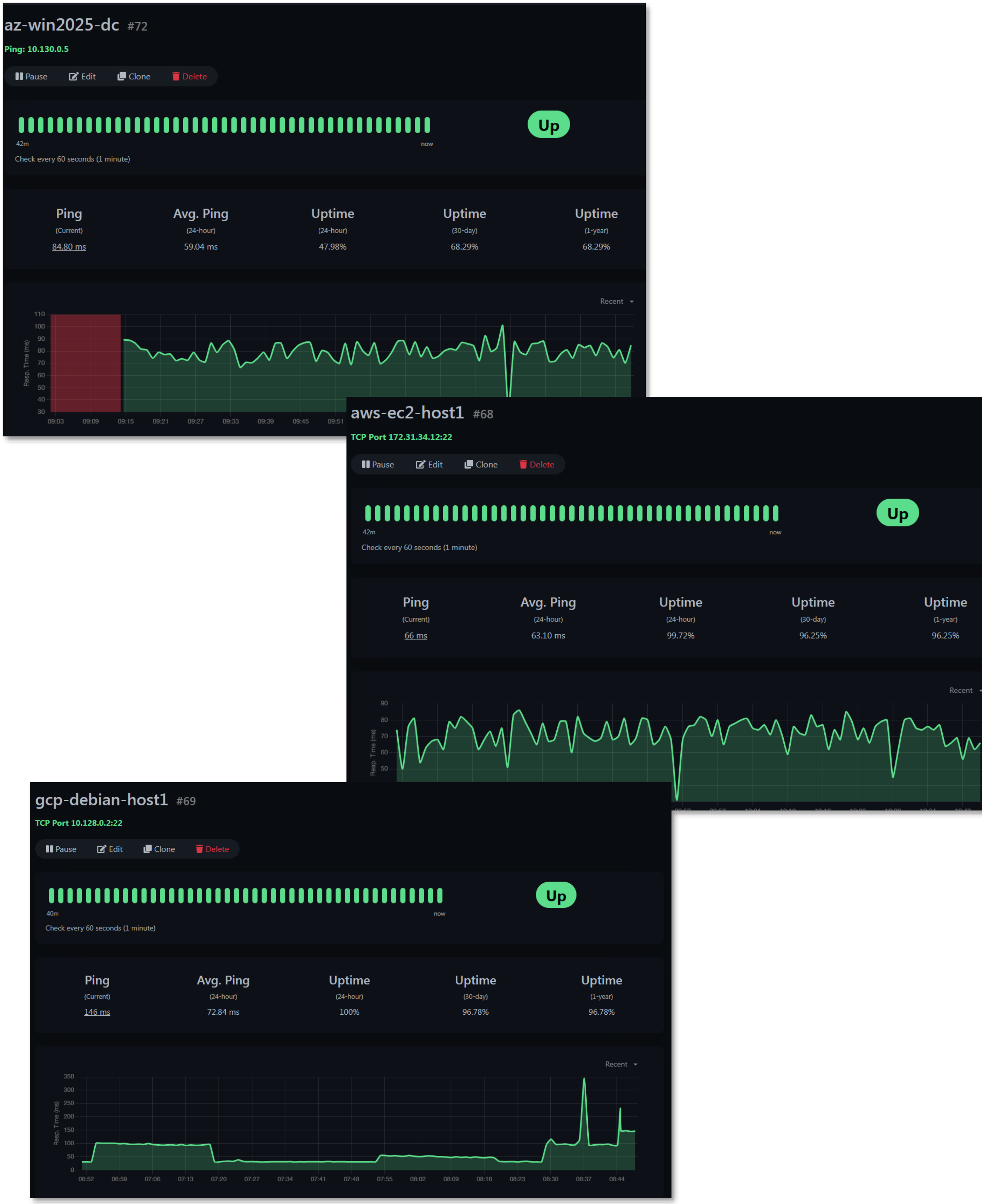
Playbook	Function	Target Cloud Nodes
new_install_baseline_roles.yml	Bootstrap: Wazuh agent, SSH hardening, user accounts, Checkmk agent, base packages	All Linux cloud nodes (first provision)
linux_hardening.yml	SSH hardening, sysctl parameters, login banner, UFW/firewalld rules	All Linux cloud nodes (recurring)
update_linux_hosts.yml	OS package updates (apt dist-upgrade / dnf update) with reboot detection	All Linux cloud nodes (weekly, n8n scheduled)
user_mgmt.yml	ansible/paul user credentials, SSH key distribution	All Linux cloud nodes (quarterly rotation)
windows_baseline.yml	Windows hardening baseline, Wazuh agent install, Checkmk agent install	AWS Win2025, Azure Win2025 (first provision)

Uptime Kuma

Uptime Kuma (192.168.1.181) performs Ping and TCP health checks against all cloud nodes via the Tailscale tunnel, validating both service availability and tunnel connectivity. Failures trigger Discord webhook notifications to the #uptime-kuma channel within the alert latency target.



Monitor	Check Type	Target	Alert Channel
aws-ec2-host1 tunnel	TCP port	172.31.34.12:22	#uptime-kuma (Discord)
aws-win2025-host2 tunnel	ping	172.31.46.238	#uptime-kuma (Discord)
gcp-debian-host1 tunnel	TCP port	100.67.103.120:22	#uptime-kuma (Discord)
az-ubuntu availability	TCP port	10.130.0.7:22	#uptime-kuma (Discord)
az-win2025-dc tunnel	ping	10.130.0.5	#uptime-kuma (Discord)


az-win2025-dc #72
Ping: 10.130.0.5
Pause
Edit
Clone
Delete
Up
42m
now
Check every 60 seconds (1 minute)
Ping (Current) 84.80 ms
Avg. Ping (24-hour) 59.04 ms
Uptime (24-hour) 47.98%
Uptime (30-day) 68.29%
Uptime (1-year) 68.29%
Recent
Resp. Time (ms)
aws-ec2-host1 #68
TCP Port 172.31.34.12:22
Pause
Edit
Clone
Delete
Up
42m
now
Check every 60 seconds (1 minute)
Ping (Current) 66 ms
Avg. Ping (24-hour) 63.10 ms
Uptime (24-hour) 99.72%
Uptime (30-day) 96.25%
Uptime (1-year) 96.25%
Recent
Resp. Time (ms)
gcp-debian-host1 #69
TCP Port 10.128.0.2:22
Pause
Edit
Clone
Delete
Up
40m
now
Check every 60 seconds (1 minute)
Ping (Current) 146 ms
Avg. Ping (24-hour) 72.84 ms
Uptime (24-hour) 100%
Uptime (30-day) 96.78%
Uptime (1-year) 96.78%
Recent
Resp. Time (ms)